

In [1]:

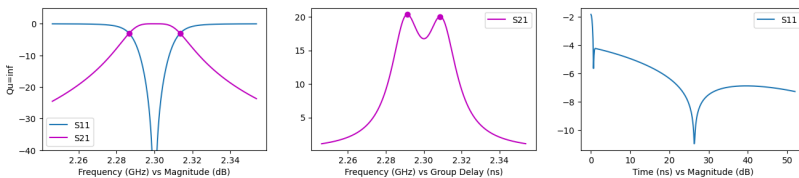
```
import numpy as np
from catalog import draw, COUPLED, LOWPASS
```

In [2]:

```
fo = 2.3e9
bw = 26.9e6
# From the Electronic Filter Design Handbook, p5-21
for name, table in list(COUPLED.items()):
    for d in table:
        qk = np.concatenate((d[1:], d[2:], d[1:2]))
        print('{} Pole {}'.format(len(qk)-1, name))
        draw(qk, bw, fo)
```

2 Pole Butterworth

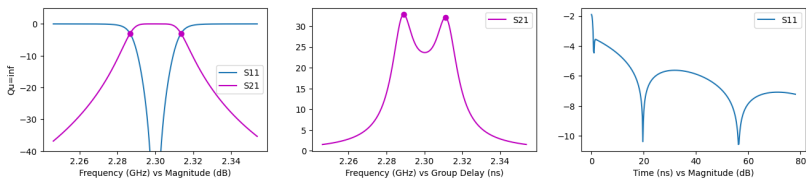
3dB Bandwidth = 26.89193 MHz
Delay Bandwidth = 17.30381 MHz
Transmission Delay = 16.73703 ns
Minimum Return Loss = 70.399 dB
Insertion Loss = 0.000 dB
[1.414 0.707 1.414]



3 Pole Butterworth

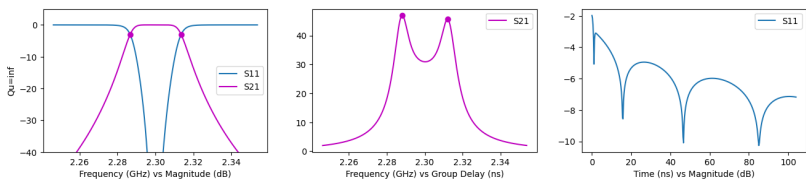
3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 21.98488 MHz
Transmission Delay = 23.66974 ns
Minimum Return Loss = 266.066 dB

Insertion Loss = 0.000 dB
[1. 0.707 0.707 1.]



4 Pole Butterworth

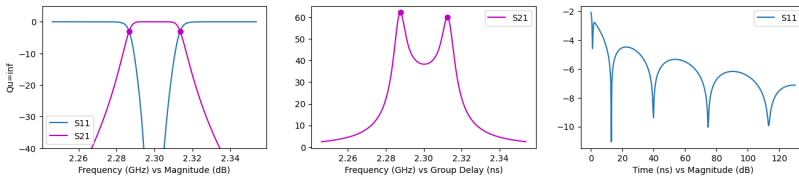
3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 23.87883 MHz
Transmission Delay = 30.92375 ns
Minimum Return Loss = 69.804 dB
Insertion Loss = 0.000 dB
[0.765 0.841 0.541 0.841 0.765]



5 Pole Butterworth

3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 24.84732 MHz

Transmission Delay = 38.28810 ns
Minimum Return Loss = 86.770 dB
Insertion Loss = 0.000 dB
[0.618 1. 0.556 0.556 1. 0.618]



6 Pole Butterworth

3dB Bandwidth = 26.91345 MHz

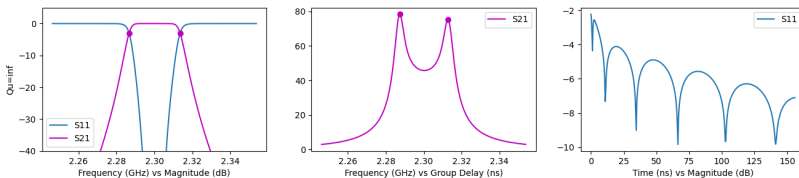
Delay Bandwidth = 25.40690 MHz

Transmission Delay = 45.70850 ns

Minimum Return Loss = 52.534 dB

Insertion Loss = 0.000 dB

[0.518 1.169 0.605 0.518 0.605 1.169 0.518]



7 Pole Butterworth

3dB Bandwidth = 26.91345 MHz

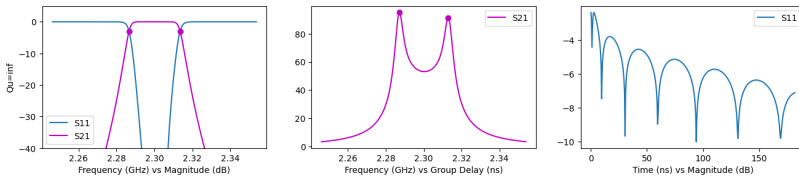
Delay Bandwidth = 25.76202 MHz

Transmission Delay = 53.17285 ns

Minimum Return Loss = 81.024 dB

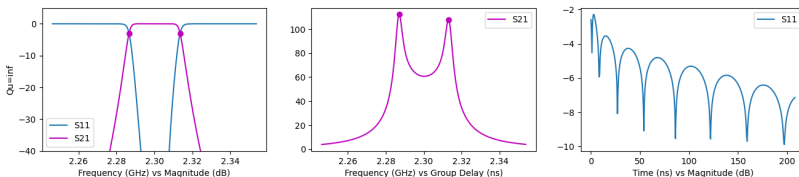
Insertion Loss = 0.000 dB

[0.445 1.342 0.667 0.527 0.527 0.667 1.342 0.445]



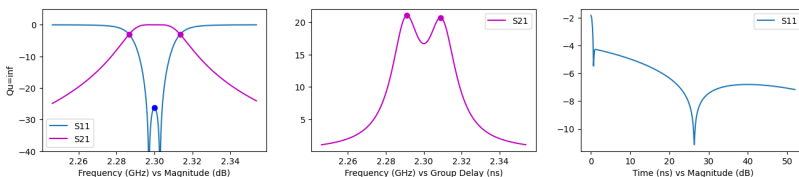
8 Pole Butterworth

3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 26.00952 MHz
 Transmission Delay = 60.62724 ns
 Minimum Return Loss = 65.272 dB
 Insertion Loss = 0.000 dB
 [0.39 1.519 0.736 0.554 0.51 0.554 0.736 1.519 0.39]



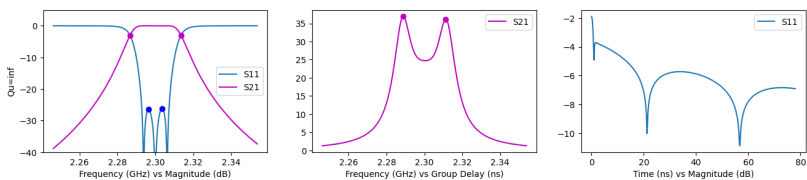
2 Pole Chebyshev 0.01 dB

3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 17.60512 MHz
 Transmission Delay = 16.69355 ns
 Minimum Return Loss = 26.245 dB
 Insertion Loss = 0.000 dB
 [1.483 0.708 1.483]



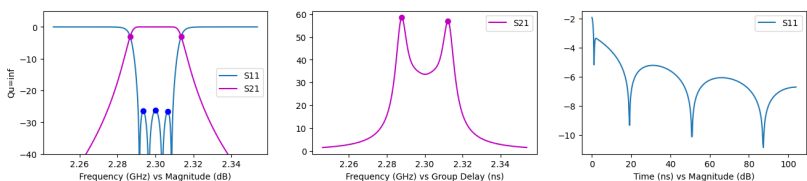
3 Pole Chebyshev 0.01 dB

3dB Bandwidth = 26.91345 MHz
Delay Bandwidth = 22.42608 MHz
Transmission Delay = 24.74570 ns
Minimum Return Loss = 26.361 dB
Insertion Loss = 0.000 dB
[1.181 0.682 0.682 1.181]



4 Pole Chebyshev 0.01 dB

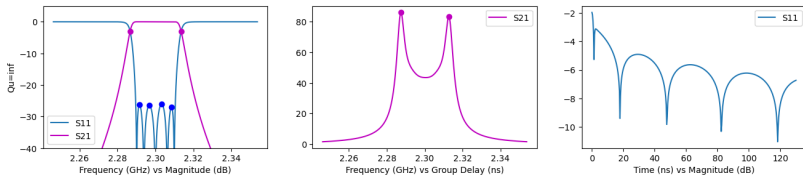
3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 24.30927 MHz
Transmission Delay = 33.61811 ns
Minimum Return Loss = 26.447 dB
Insertion Loss = 0.000 dB
[1.046 0.737 0.541 0.737 1.046]



5 Pole Chebyshev 0.01 dB

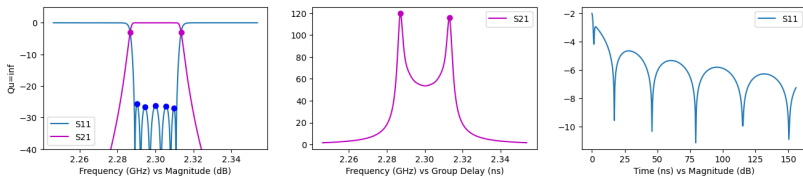
3dB Bandwidth = 26.91345 MHz
Delay Bandwidth = 25.25625 MHz
Transmission Delay = 43.52876 ns

Minimum Return Loss = 26.281 dB
Insertion Loss = 0.000 dB
[0.977 0.78 0.54 0.54 0.78 0.977]



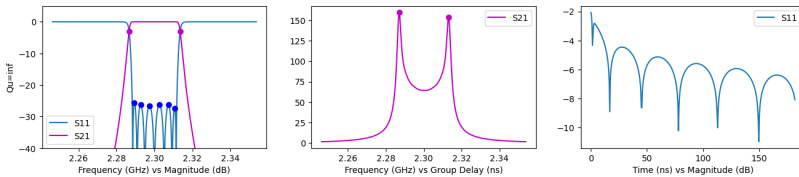
6 Pole Chebyshev 0.01 dB

3dB Bandwidth = 26.91345 MHz
Delay Bandwidth = 25.75126 MHz
Transmission Delay = 53.60095 ns
Minimum Return Loss = 26.345 dB
Insertion Loss = 0.000 dB
[0.937 0.809 0.55 0.518 0.55 0.809 0.937]



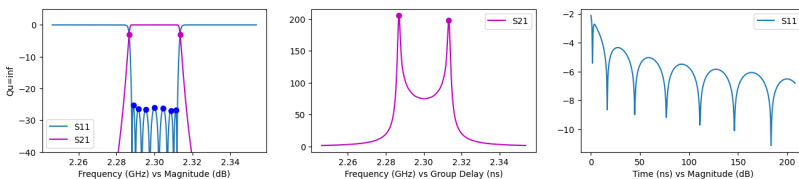
7 Pole Chebyshev 0.01 dB

3dB Bandwidth = 26.92421 MHz
Delay Bandwidth = 26.07409 MHz
Transmission Delay = 64.40146 ns
Minimum Return Loss = 26.253 dB
Insertion Loss = 0.000 dB
[0.913 0.829 0.56 0.517 0.517 0.56 0.829 0.913]



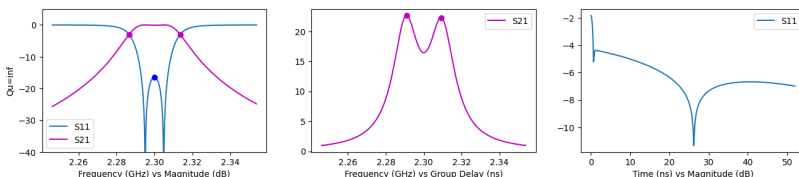
8 Pole Chebyshev 0.01 dB

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 26.25703 MHz
 Transmission Delay = 75.03809 ns
 Minimum Return Loss = 26.418 dB
 Insertion Loss = 0.000 dB
 [0.897 0.843 0.567 0.52 0.51 0.52 0.567 0.843 0.897]



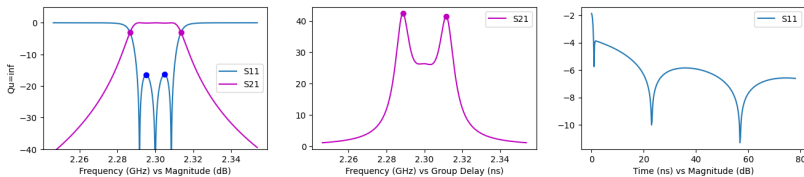
2 Pole Chebyshev 0.1 dB

3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 18.13241 MHz
 Transmission Delay = 16.45147 ns
 Minimum Return Loss = 16.407 dB
 Insertion Loss = 0.000 dB
 [1.638 0.711 1.638]



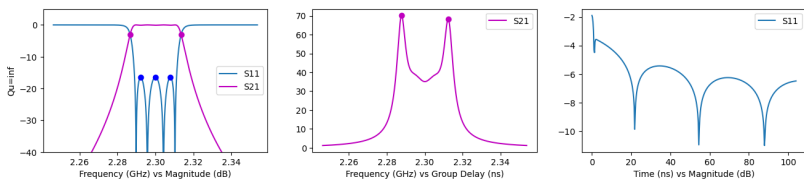
3 Pole Chebyshev 0.1 dB

3dB Bandwidth = 26.91345 MHz
Delay Bandwidth = 22.83500 MHz
Transmission Delay = 26.37799 ns
Minimum Return Loss = 16.408 dB
Insertion Loss = 0.000 dB
[1.433 0.662 0.662 1.433]



4 Pole Chebyshev 0.1 dB

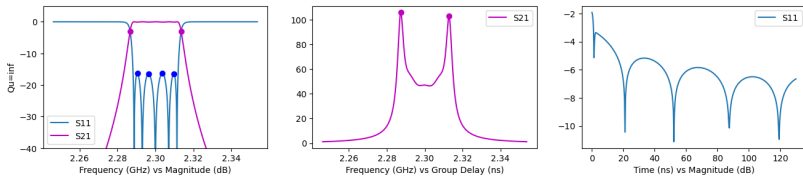
3dB Bandwidth = 26.89193 MHz
Delay Bandwidth = 24.58906 MHz
Transmission Delay = 35.09330 ns
Minimum Return Loss = 16.418 dB
Insertion Loss = 0.000 dB
[1.345 0.685 0.542 0.685 1.345]



5 Pole Chebyshev 0.1 dB

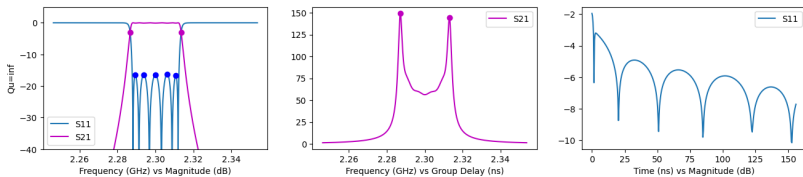
3dB Bandwidth = 26.92421 MHz
Delay Bandwidth = 25.43918 MHz
Transmission Delay = 47.03990 ns

Minimum Return Loss = 16.375 dB
 Insertion Loss = 0.000 dB
 [1.301 0.703 0.536 0.536 0.703 1.301]



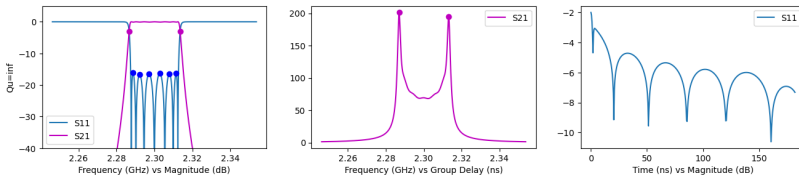
6 Pole Chebyshev 0.1 dB

3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 25.89115 MHz
 Transmission Delay = 56.27378 ns
 Minimum Return Loss = 16.424 dB
 Insertion Loss = 0.000 dB
 [1.277 0.715 0.539 0.518 0.539 0.715 1.277]



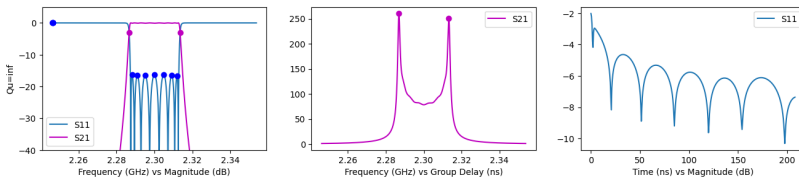
7 Pole Chebyshev 0.1 dB

3dB Bandwidth = 26.91345 MHz
 Delay Bandwidth = 26.16018 MHz
 Transmission Delay = 69.34257 ns
 Minimum Return Loss = 16.407 dB
 Insertion Loss = 0.000 dB
 [1.262 0.722 0.542 0.516 0.516 0.542 0.722 1.262]



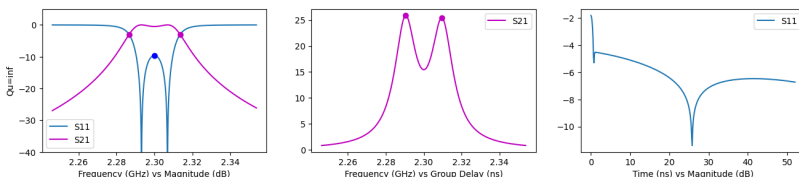
8 Pole Chebyshev 0.1 dB

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 26.32159 MHz
 Transmission Delay = 78.44892 ns
 Minimum Return Loss = 16.349 dB
 Insertion Loss = 0.000 dB
 [1.251 0.728 0.545 0.516 0.51 0.516 0.545 0.728 1.251]



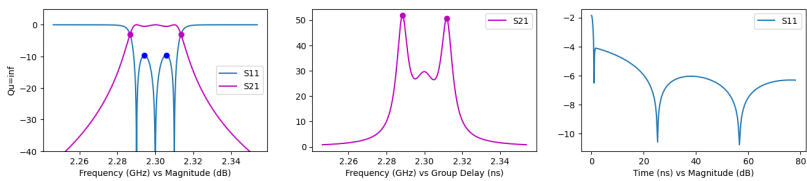
2 Pole Chebyshev 0.5 dB

3dB Bandwidth = 26.91345 MHz
 Delay Bandwidth = 18.95026 MHz
 Transmission Delay = 15.44645 ns
 Minimum Return Loss = 9.614 dB
 Insertion Loss = 0.000 dB
 [1.95 0.723 1.95]



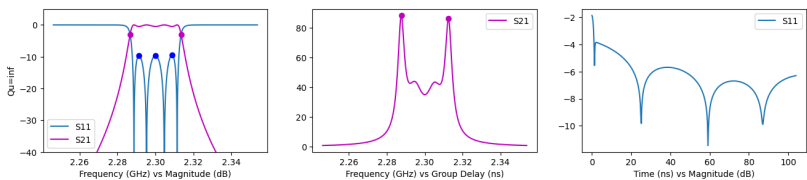
3 Pole Chebyshev 0.5 dB

3dB Bandwidth = 26.88117 MHz
Delay Bandwidth = 23.28697 MHz
Transmission Delay = 29.63938 ns
Minimum Return Loss = 9.643 dB
Insertion Loss = 0.000 dB
[1.864 0.647 0.647 1.864]



4 Pole Chebyshev 0.5 dB

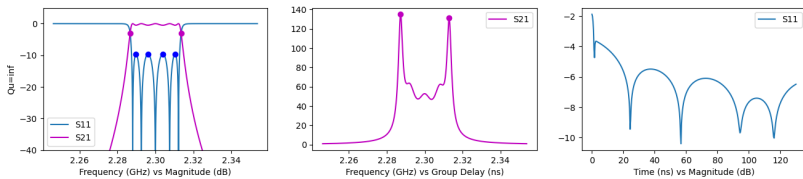
3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 24.86885 MHz
Transmission Delay = 35.01060 ns
Minimum Return Loss = 9.661 dB
Insertion Loss = 0.000 dB
[1.826 0.648 0.545 0.648 1.826]



5 Pole Chebyshev 0.5 dB

3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 25.61136 MHz
Transmission Delay = 52.72494 ns

Minimum Return Loss = 9.638 dB
 Insertion Loss = 0.000 dB
 [1.807 0.652 0.534 0.534 0.652 1.807]



6 Pole Chebyshev 0.5 dB

3dB Bandwidth = 26.91345 MHz

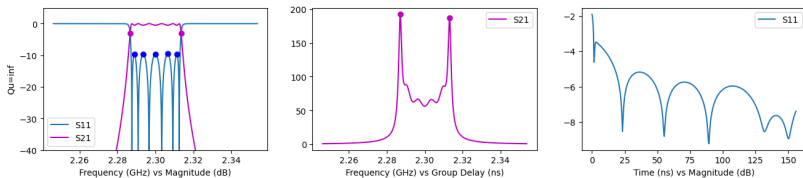
Delay Bandwidth = 26.00952 MHz

Transmission Delay = 56.19745 ns

Minimum Return Loss = 9.651 dB

Insertion Loss = 0.000 dB

[1.796 0.655 0.533 0.519 0.533 0.655 1.796]



7 Pole Chebyshev 0.5 dB

3dB Bandwidth = 26.92421 MHz

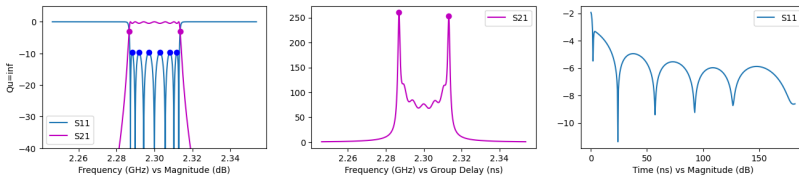
Delay Bandwidth = 26.25703 MHz

Transmission Delay = 76.84949 ns

Minimum Return Loss = 9.601 dB

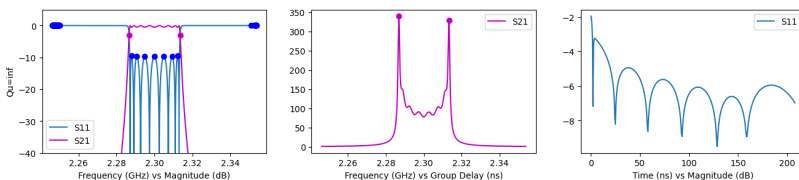
Insertion Loss = 0.000 dB

[1.79 0.657 0.533 0.516 0.516 0.533 0.657 1.79]



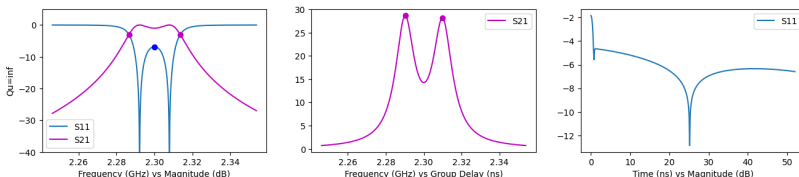
8 Pole Chebyshev 0.5 dB

3dB Bandwidth = 26.91345 MHz
 Delay Bandwidth = 26.40768 MHz
 Transmission Delay = 77.85405 ns
 Minimum Return Loss = 0.000 dB
 Insertion Loss = 0.000 dB
 [1.785 0.658 0.533 0.515 0.511 0.515 0.533 0.658 1.785]



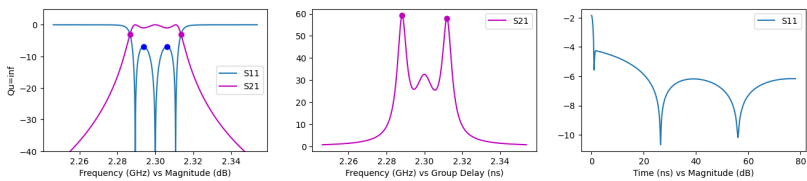
2 Pole Chebyshev 1.0 dB

3dB Bandwidth = 27.04258 MHz
 Delay Bandwidth = 19.57440 MHz
 Transmission Delay = 14.26174 ns
 Minimum Return Loss = 6.847 dB
 Insertion Loss = 0.000 dB
 [2.21 0.739 2.21]



3 Pole Chebyshev 1.0 dB

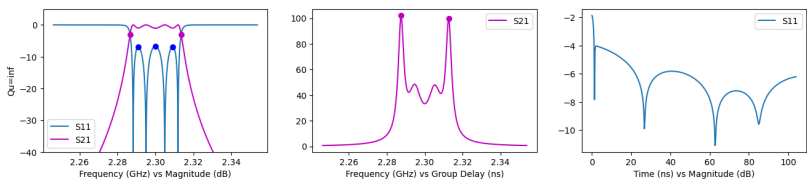
3dB Bandwidth = 26.94573 MHz
Delay Bandwidth = 23.63132 MHz
Transmission Delay = 32.58622 ns
Minimum Return Loss = 6.881 dB
Insertion Loss = 0.000 dB
[2.21 0.645 0.645 2.21]



4 Pole Chebyshev 1.0 dB

3dB Bandwidth = 26.94573 MHz
Delay Bandwidth = 25.08407 MHz

Transmission Delay = 33.30642 ns
Minimum Return Loss = 6.900 dB
Insertion Loss = 0.000 dB
[2.21 0.638 0.546 0.638 2.21]

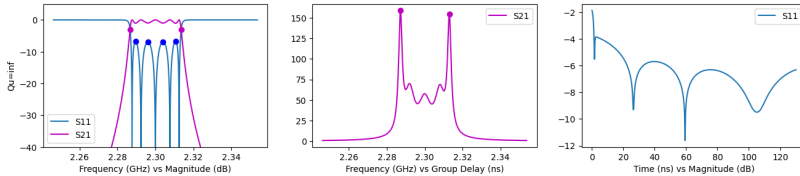


5 Pole Chebyshev 1.0 dB

3dB Bandwidth = 26.95650 MHz

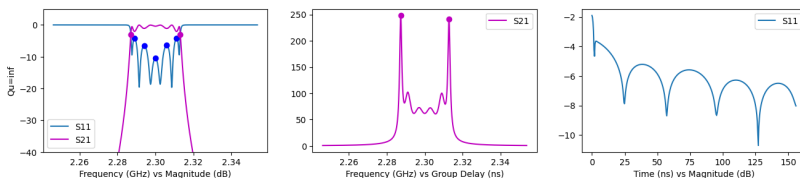
Delay Bandwidth = 25.76202 MHz

Transmission Delay = 57.71616 ns
 Minimum Return Loss = 6.812 dB
 Insertion Loss = 0.000 dB
 [2.21 0.633 0.535 0.538 0.633 2.21]



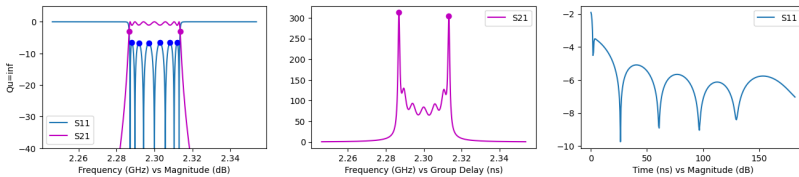
6 Pole Chebyshev 1.0 dB

 3dB Bandwidth = 26.12789 MHz
 Delay Bandwidth = 25.44994 MHz
 Transmission Delay = 61.80017 ns
 Minimum Return Loss = 6.422 dB
 Insertion Loss = 0.000 dB
 [2.25 0.631 0.531 0.51 0.531 0.531 2.25]



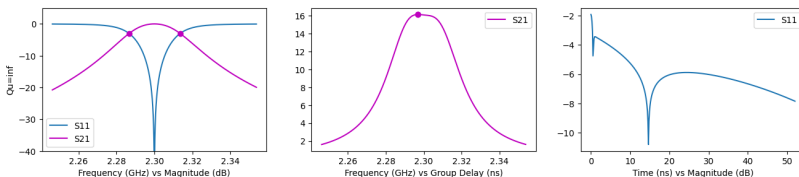
7 Pole Chebyshev 1.0 dB

 3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 26.32159 MHz
 Transmission Delay = 84.51237 ns
 Minimum Return Loss = 6.632 dB
 Insertion Loss = 0.000 dB
 [2.25 0.631 0.53 0.517 0.517 0.53 0.631 2.25]



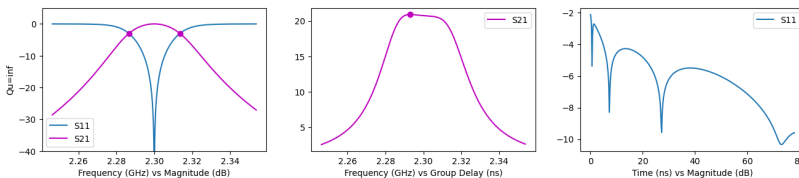
2 Pole Bessel

3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 0.00000 MHz
 Transmission Delay = 16.10322 ns
 Minimum Return Loss = 63.741 dB
 Insertion Loss = 0.000 dB
 [0.5755 0.9 2.148]



3 Pole Bessel

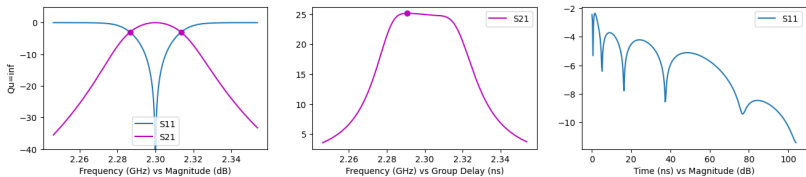
3dB Bandwidth = 26.91345 MHz
 Delay Bandwidth = 0.00000 MHz
 Transmission Delay = 20.76587 ns
 Minimum Return Loss = 66.437 dB
 Insertion Loss = 0.000 dB
 [0.337 1.748 0.684 2.203]



4 Pole Bessel

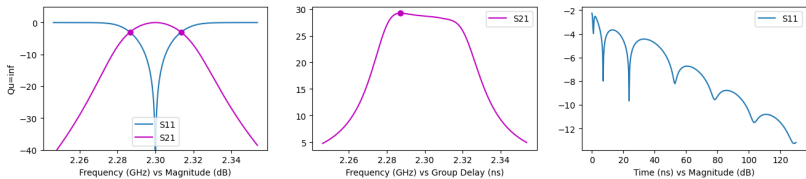
3dB Bandwidth = 26.94573 MHz
Delay Bandwidth = 0.09685 MHz
Transmission Delay = 24.95972 ns

Minimum Return Loss = 55.020 dB
Insertion Loss = 0.000 dB
[0.233 2.53 1.175 0.644 2.24]



5 Pole Bessel

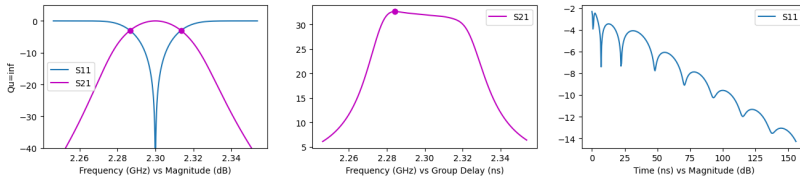
3dB Bandwidth = 26.95650 MHz
Delay Bandwidth = 0.02152 MHz
Transmission Delay = 28.69082 ns
Minimum Return Loss = 50.953 dB
Insertion Loss = 0.000 dB
[0.394 1.91 0.75 0.65 1.987 0.275]



6 Pole Bessel

3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 0.00000 MHz

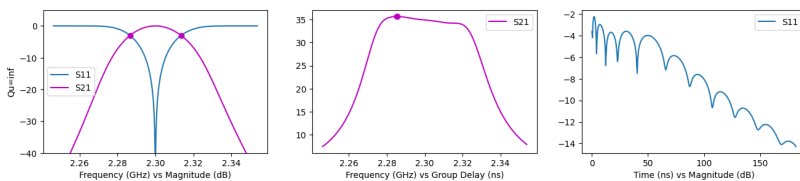
Transmission Delay = 31.97133 ns
 Minimum Return Loss = 59.404 dB
 Insertion Loss = 0.000 dB
 [0.415 2. 0.811 0.601 1.253 3.038 0.187]



7 Pole Bessel

3dB Bandwidth = 26.93497 MHz
 Delay Bandwidth = 0.03228 MHz
 Transmission Delay = 34.92021 ns

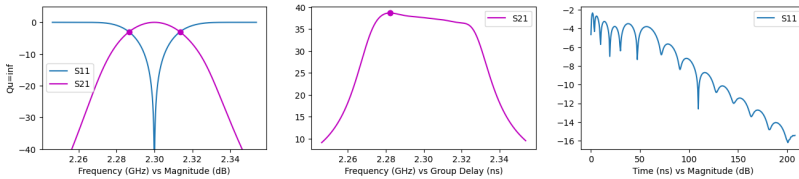
Minimum Return Loss = 53.148 dB
 Insertion Loss = 0.000 dB
 [0.187 3.325 1.66 1.293 0.695 0.674 2.203 0.242]



8 Pole Bessel

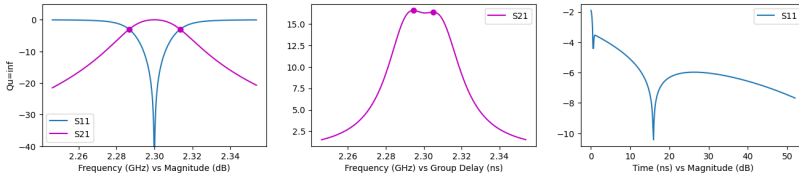
3dB Bandwidth = 26.89193 MHz
 Delay Bandwidth = 0.00000 MHz
 Transmission Delay = 37.63763 ns
 Minimum Return Loss = 60.735 dB

Insertion Loss = 0.000 dB
 [0.139 4.284 2.079 1.484 1.246 0.678 0.697 2.286 0.242]



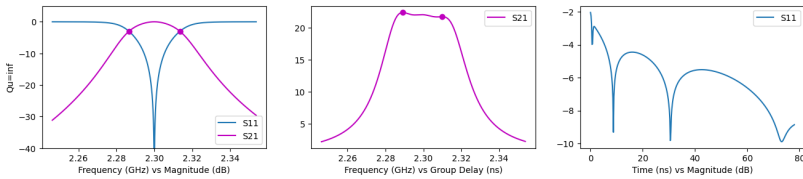
2 Pole Linear Phase 0.05 Deg

3dB Bandwidth = 26.91345 MHz
 Delay Bandwidth = 10.47053 MHz
 Transmission Delay = 16.30065 ns
 Minimum Return Loss = 63.226 dB
 Insertion Loss = 0.000 dB
 [0.648 0.856 2.109]



3 Pole Linear Phase 0.05 Deg

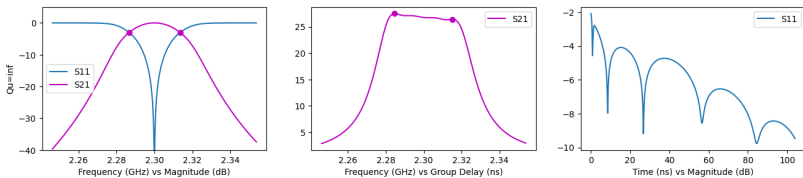
3dB Bandwidth = 26.89193 MHz
 Delay Bandwidth = 20.82268 MHz
 Transmission Delay = 22.07687 ns
 Minimum Return Loss = 60.414 dB
 Insertion Loss = 0.000 dB
 [0.433 1.489 0.652 2.254]



4 Pole Linear Phase 0.05 Deg

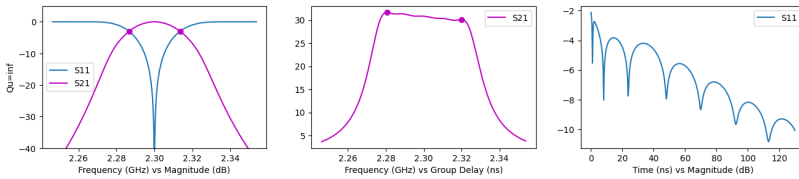
3dB Bandwidth = 26.89193 MHz
 Delay Bandwidth = 30.40004 MHz
 Transmission Delay = 26.75177 ns
 Minimum Return Loss = 63.995 dB

Insertion Loss = 0.000 dB
 [0.493 1.632 0.718 0.739 0.718]



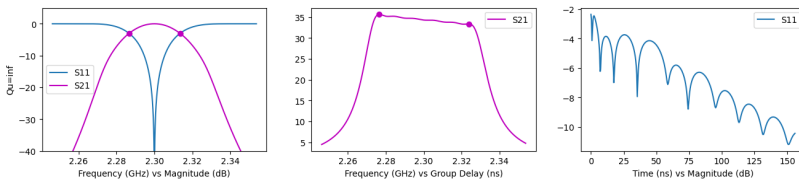
5 Pole Linear Phase 0.05 Deg

3dB Bandwidth = 26.88117 MHz
 Delay Bandwidth = 39.37478 MHz
 Transmission Delay = 30.83116 ns
 Minimum Return Loss = 64.402 dB
 Insertion Loss = 0.000 dB
 [0.547 1.8 0.848 0.584 1.372 0.446]



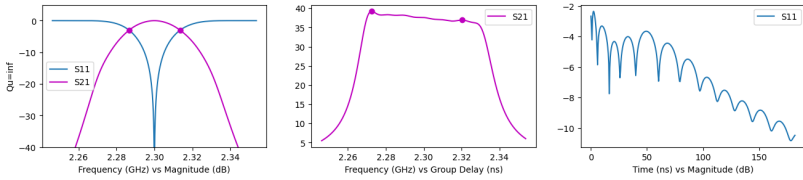
6 Pole Linear Phase 0.05 Deg

3dB Bandwidth	=	26.90269 MHz
Delay Bandwidth	=	47.14427 MHz
Transmission Delay	=	34.30793 ns
Minimum Return Loss	=	69.345 dB
Insertion Loss	=	0.000 dB
[0.397 1.993 1.379 0.683 0.661 1.553 0.468]		



7 Pole Linear Phase 0.05 Deg

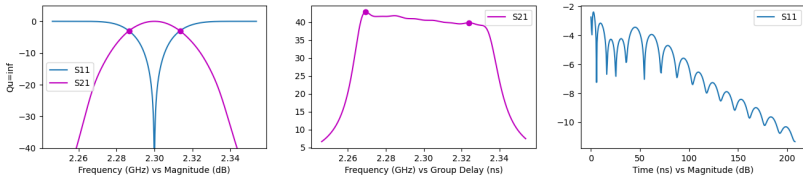
3dB Bandwidth	=	26.89193 MHz
Delay Bandwidth	=	47.58548 MHz
Transmission Delay	=	37.59280 ns
Minimum Return Loss	=	78.515 dB
Insertion Loss	=	0.000 dB
[0.316 2.49 1.442 1.446 0.927 0.579 1.26 0.484]		



8 Pole Linear Phase 0.05 Deg

3dB Bandwidth	=	26.93497 MHz
Delay Bandwidth	=	54.50485 MHz
Transmission Delay	=	40.54840 ns
Minimum Return Loss	=	55.048 dB

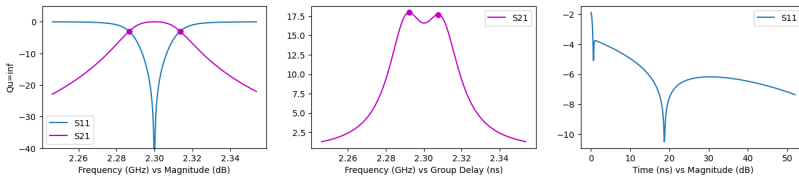
Insertion Loss = 0.000 dB
 [0.335 2.585 1.484 1.602 1.16 0.596 0.868 1.733 0.363]



2 Pole Linear Phase 0.5 Deg

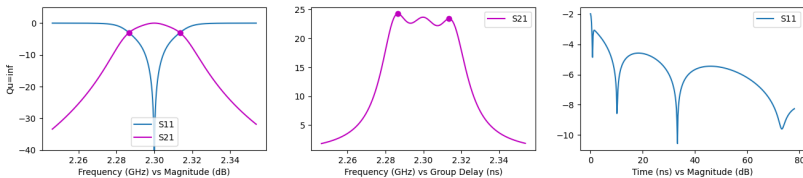
3dB Bandwidth	=	26.92421 MHz
Delay Bandwidth	=	15.24844 MHz
Transmission Delay	=	16.58362 ns
Minimum Return Loss	=	62.617 dB
Insertion Loss	=	0.000 dB

[0.825 0.783 1.98]



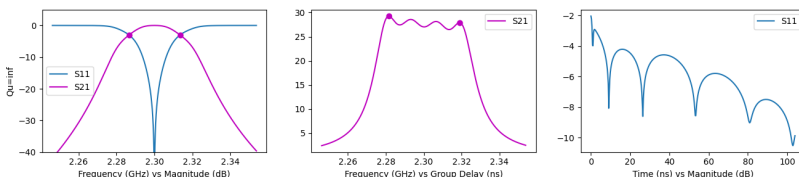
3 Pole Linear Phase 0.5 Deg

3dB Bandwidth = 26.89193 MHz
 Delay Bandwidth = 27.04258 MHz
 Transmission Delay = 23.67118 ns
 Minimum Return Loss = 74.207 dB
 Insertion Loss = 0.000 dB
 [0.553 1.33 0.635 2.425]



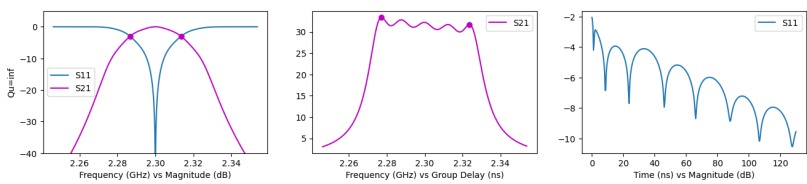
4 Pole Linear Phase 0.5 Deg

3dB Bandwidth = 26.89193 MHz
 Delay Bandwidth = 37.26561 MHz
 Transmission Delay = 27.02043 ns
 Minimum Return Loss = 60.983 dB
 Insertion Loss = 0.000 dB
 [0.581 1.575 0.797 0.656 1.026]



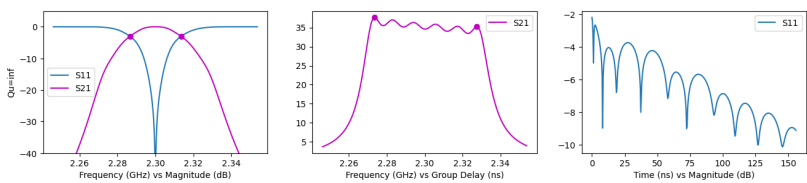
5 Pole Linear Phase 0.5 Deg

3dB Bandwidth = 26.88117 MHz
Delay Bandwidth = 46.36948 MHz
Transmission Delay = 32.24601 ns
Minimum Return Loss = 69.819 dB
Insertion Loss = 0.000 dB
[0.664 1.779 0.919 0.576 1.162 0.611]



6 Pole Linear Phase 0.5 Deg

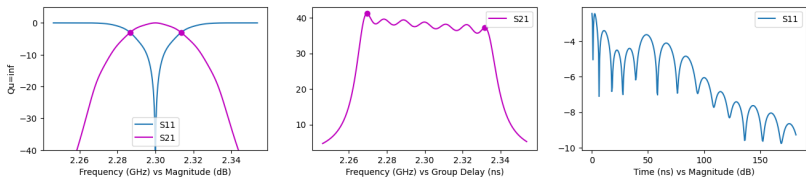
3dB Bandwidth = 26.92421 MHz
Delay Bandwidth = 53.91299 MHz
Transmission Delay = 34.60845 ns
Minimum Return Loss = 63.319 dB
Insertion Loss = 0.000 dB
[0.552 1.874 1.355 0.641 0.721 1.429 0.586]



7 Pole Linear Phase 0.5 Deg

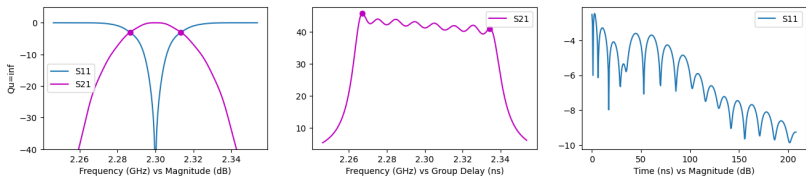
3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 62.23130 MHz
Transmission Delay = 38.76800 ns

Minimum Return Loss = 59.825 dB
Insertion Loss = 0.000 dB
[0.401 2.324 1.394 1.5 1.079 0.59 1.045 0.688]



8 Pole Linear Phase 0.5 Deg

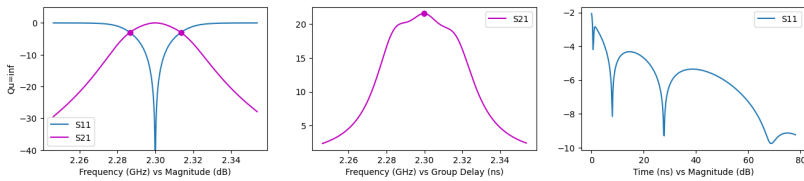
3dB Bandwidth = 26.85965 MHz
Delay Bandwidth = 67.24596 MHz
Transmission Delay = 41.16286 ns
Minimum Return Loss = 62.499 dB
Insertion Loss = 0.000 dB
[0.415 2.41 1.47 1.527 1.409 0.659 0.755 1.335 0.563]



3 Pole Gaussian 6 dB

3dB Bandwidth = 26.89193 MHz
Delay Bandwidth = 0.00000 MHz
Transmission Delay = 21.52194 ns
Minimum Return Loss = 74.847 dB

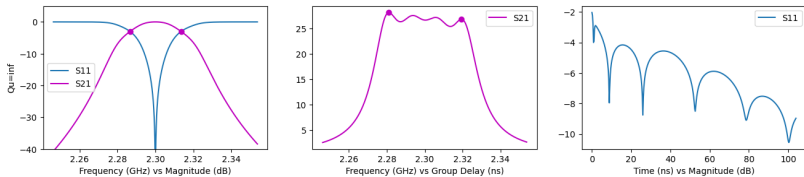
Insertion Loss = 0.000 dB
 [0.404 1.662 0.691 2.338]



4 Pole Gaussian 6 dB

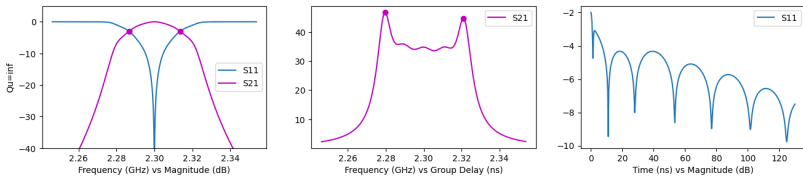
3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 38.26639 MHz
 Transmission Delay = 26.64859 ns

Minimum Return Loss = 58.595 dB
 Insertion Loss = 0.000 dB
 [0.57 1.623 0.798 0.682 0.914]



5 Pole Gaussian 6 dB

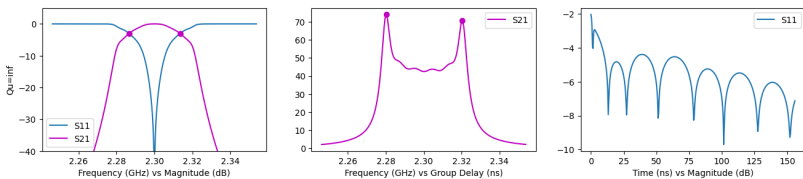
3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 41.30101 MHz
 Transmission Delay = 34.79662 ns
 Minimum Return Loss = 64.544 dB
 Insertion Loss = 0.000 dB
 [0.891 1.418 0.864 0.553 1.046 0.67]



6 Pole Gaussian 6 dB

3dB Bandwidth	=	26.93497 MHz
Delay Bandwidth	=	40.11729 MHz
Transmission Delay	=	42.47878 ns
Minimum Return Loss	=	56.914 dB

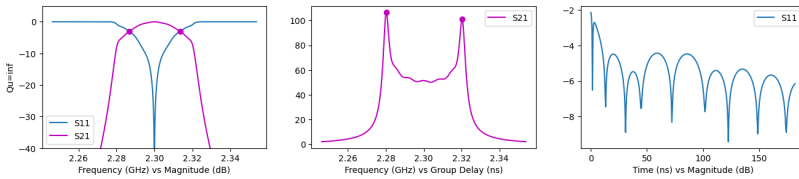
Insertion Loss = 0.000 dB
 [0.883 1.172 1.029 0.595 0.605 1.094 0.752]



7 Pole Gaussian 6 dB

3dB Bandwidth	=	26.89193 MHz
Delay Bandwidth	=	39.95588 MHz
Transmission Delay	=	51.53573 ns

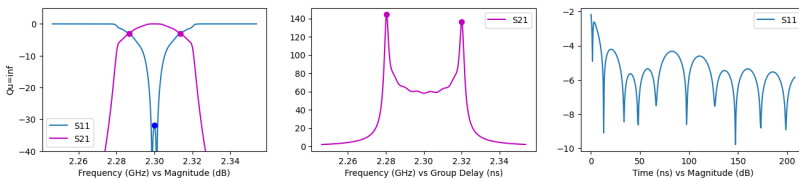
Minimum Return Loss = 61.118 dB
 Insertion Loss = 0.000 dB
 [0.736 1.13 0.955 0.884 0.534 0.633 1.104 0.93]



8 Pole Gaussian 6 dB

3dB Bandwidth	=	26.89193 MHz
Delay Bandwidth	=	39.65457 MHz
Transmission Delay	=	58.15587 ns
Minimum Return Loss	=	31.908 dB

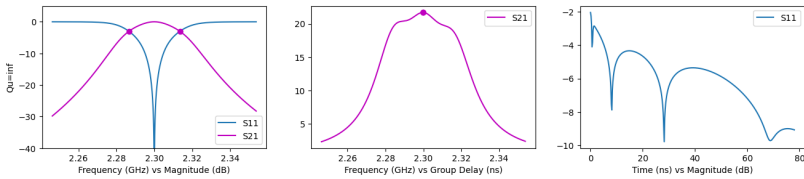
Insertion Loss = 0.000 dB
 [0.738 1.124 0.866 0.922 0.708 0.501 0.752 1.089 0.948]



3 Pole Gaussian 12 dB

3dB Bandwidth	=	26.89193 MHz
Delay Bandwidth	=	0.00000 MHz
Transmission Delay	=	21.69219 ns
Minimum Return Loss	=	74.430 dB
Insertion Loss	=	0.000 dB

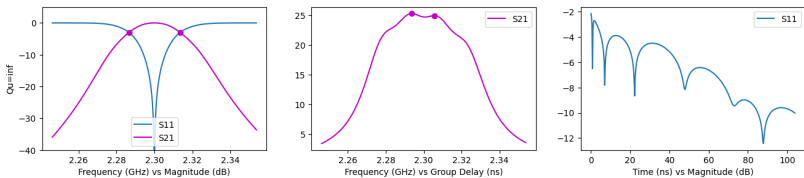
[0.415 1.631 0.686 2.345]



4 Pole Gaussian 12 dB

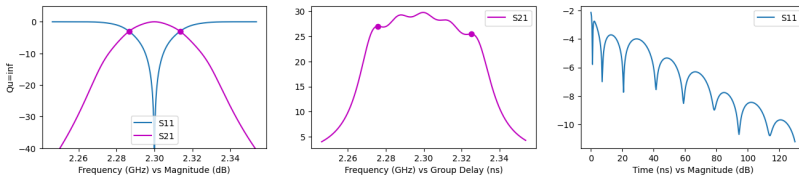
3dB Bandwidth = 26.91345 MHz
 Delay Bandwidth = 11.89099 MHz
 Transmission Delay = 24.69637 ns

Minimum Return Loss = 59.830 dB
 Insertion Loss = 0.000 dB
 [0.419 1.989 0.833 0.74 0.766]



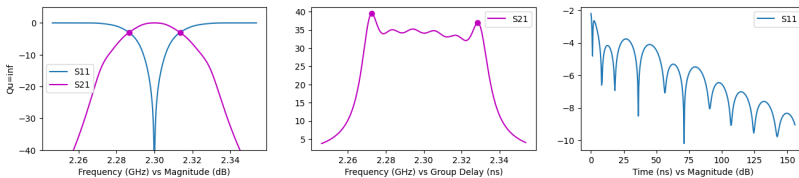
5 Pole Gaussian 12 dB

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 49.18888 MHz
 Transmission Delay = 29.74097 ns
 Minimum Return Loss = 60.048 dB
 Insertion Loss = 0.000 dB
 [0.534 2.085 0.976 0.605 1.333 0.503]



6 Pole Gaussian 12 dB

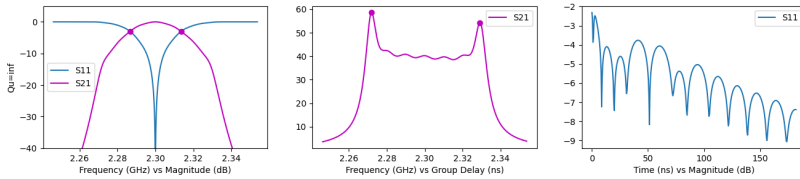
3dB Bandwidth = 26.91345 MHz
 Delay Bandwidth = 55.90379 MHz
 Transmission Delay = 34.11640 ns
 Minimum Return Loss = 81.688 dB
 Insertion Loss = 0.000 dB
 [0.543 1.839 1.442 0.686 0.707 1.468 0.558]



7 Pole Gaussian 12 dB

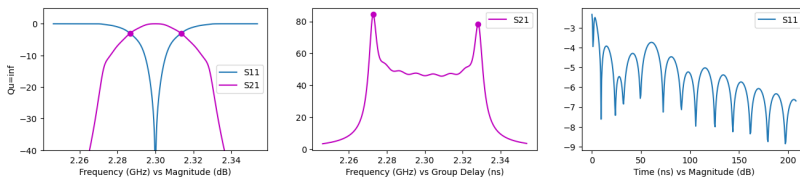
3dB Bandwidth = 26.92421 MHz
 Delay Bandwidth = 57.14131 MHz
 Transmission Delay = 40.24505 ns

Minimum Return Loss = 64.944 dB
 Insertion Loss = 0.000 dB
 [0.492 1.708 1.44 1.181 0.611 0.781 1.541 0.665]



8 Pole Gaussian 12 dB

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 55.34421 MHz
 Transmission Delay = 45.99124 ns
 Minimum Return Loss = 66.653 dB
 Insertion Loss = 0.000 dB
 [0.549 1.586 1.262 1.296 0.808 0.569 1.023 1.504 0.64]



In [3]:

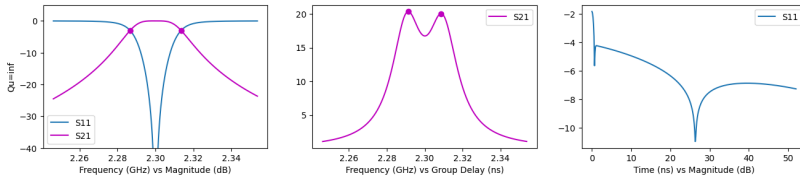
```

import ness
fo = 2.3e9
bw = 26.9e6
# from chebyshev calculator and the low pass prototype tables in zverev
for name, table in list(LOWPASS.items()):
    for d in table:
        qk = ness.coupling_g(d)
        print('{} Pole {}'.format(len(qk)-1, name))
        draw(qk, bw, fo)
  
```

2 Pole Butterworth

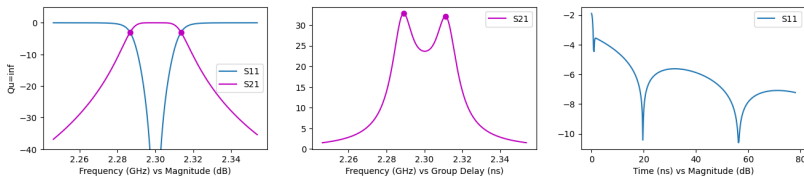
3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 17.30381 MHz

Transmission Delay = 16.73446 ns
 Minimum Return Loss = 101.377 dB
 Insertion Loss = 0.000 dB
 [1.41421 0.70710856 1.41421]



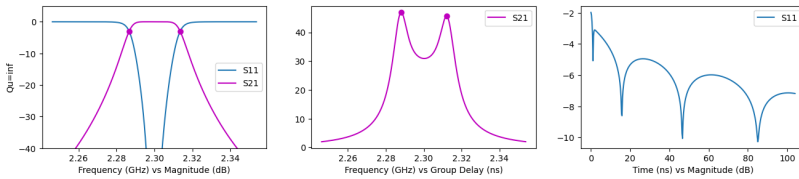
3 Pole Butterworth

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 21.98488 MHz
 Transmission Delay = 23.66617 ns
 Minimum Return Loss = 132.521 dB
 Insertion Loss = 0.000 dB
 [1. 0.70710678 0.70710678 1.]



4 Pole Butterworth

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 23.88959 MHz
 Transmission Delay = 30.92138 ns
 Minimum Return Loss = 66.555 dB
 Insertion Loss = 0.000 dB
 [0.76537 0.84089448 0.54119583 0.84089448 0.76537]



5 Pole Butterworth

3dB Bandwidth = 26.90269 MHz

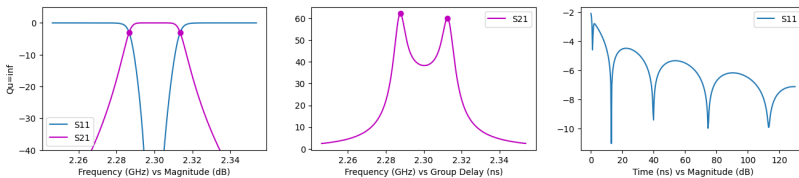
Delay Bandwidth = 24.84732 MHz

Transmission Delay = 38.29256 ns

Minimum Return Loss = 68.063 dB

Insertion Loss = 0.000 dB

[0.61803 1.00000446 0.55589366 0.55589366 1.00000446 0.61803



6 Pole Butterworth

3dB Bandwidth = 26.90269 MHz

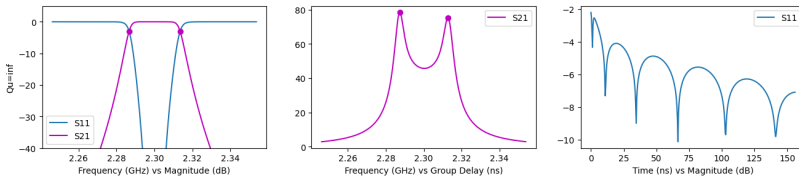
Delay Bandwidth = 25.40690 MHz

Transmission Delay = 45.71948 ns

Minimum Return Loss = 62.205 dB

Insertion Loss = 0.000 dB

[0.51764 1.16877021 0.60500135 0.51763853 0.60500135 1.16877
0.51764]



7 Pole Butterworth

3dB Bandwidth = 26.90269 MHz

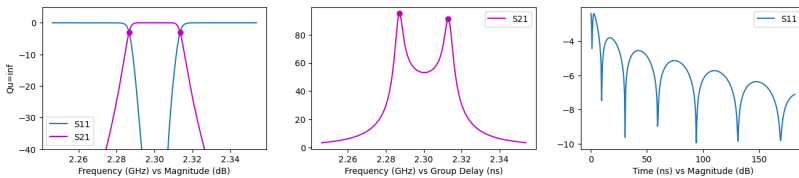
Delay Bandwidth = 25.75126 MHz

Transmission Delay = 53.17740 ns

Minimum Return Loss = 56.359 dB

Insertion Loss = 0.000 dB

[0.44504 1.34236535 0.66711406 0.52676249 0.52676249 0.66711406 1.34236535 0.44504]



8 Pole Butterworth

3dB Bandwidth = 26.90269 MHz

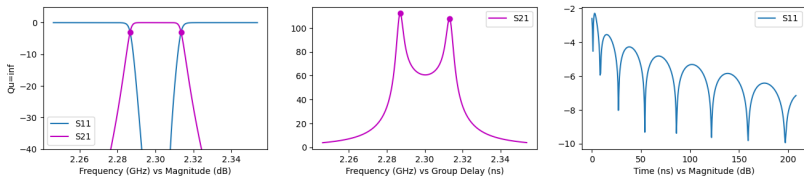
Delay Bandwidth = 25.98800 MHz

Transmission Delay = 60.65436 ns

Minimum Return Loss = 63.919 dB

Insertion Loss = 0.000 dB

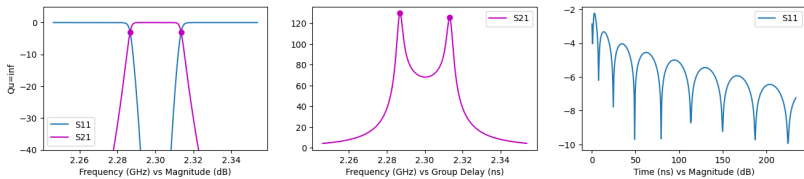
[0.39018 1.51873886 0.7356603 0.55368123 0.50979572 0.55368123 0.7356603 1.51873886 0.39018]



9 Pole Butterworth

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 26.16018 MHz
 Transmission Delay = 68.14412 ns
 Minimum Return Loss = 54.687 dB

Insertion Loss = 0.000 dB
 [0.3473 1.69686624 0.80790047 0.58931759 0.51579431 0.51579
 0.58931759 0.80790047 1.69686624 0.3473]

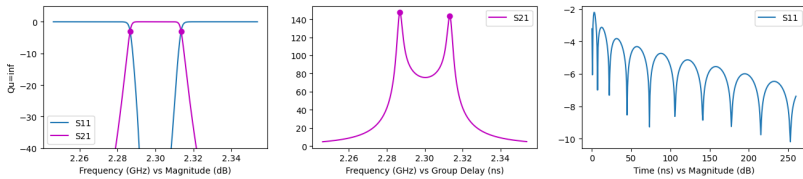


10 Pole Butterworth

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 26.28931 MHz
 Transmission Delay = 75.64239 ns

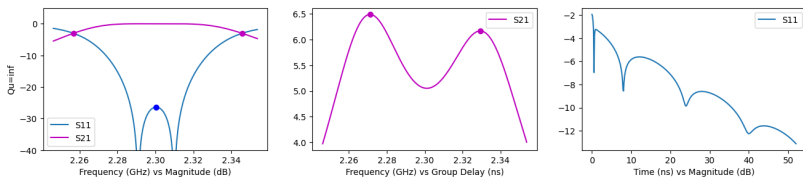
Minimum Return Loss = 65.807 dB
 Insertion Loss = 0.000 dB
 [0.31287 1.87620331 0.88248007 0.62992375 0.53299067 0.50623

0.53299067 0.62992375 0.88248007 1.87620331 0.31287]



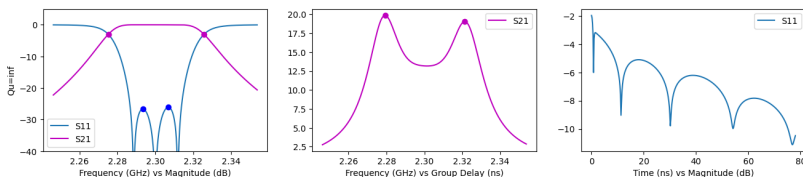
2 Pole Chebyshev 0.01 dB

3dB Bandwidth = 88.94029 MHz
Delay Bandwidth = 58.12057 MHz
Transmission Delay = 5.05693 ns
Minimum Return Loss = 26.366 dB
Insertion Loss = 0.000 dB
[0.44888 2.33728279 0.44888585]



3 Pole Chebyshev 0.01 dB

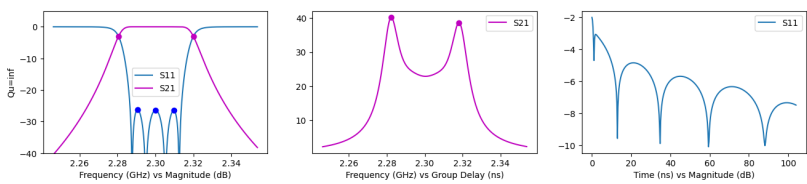
3dB Bandwidth = 50.51249 MHz
Delay Bandwidth = 42.08657 MHz
Transmission Delay = 13.18584 ns
Minimum Return Loss = 26.381 dB
Insertion Loss = 0.000 dB
[0.62918 1.27986454 1.27986454 0.62918]



4 Pole Chebyshev 0.01 dB

3dB Bandwidth = 39.47163 MHz
Delay Bandwidth = 35.68373 MHz
Transmission Delay = 22.91589 ns

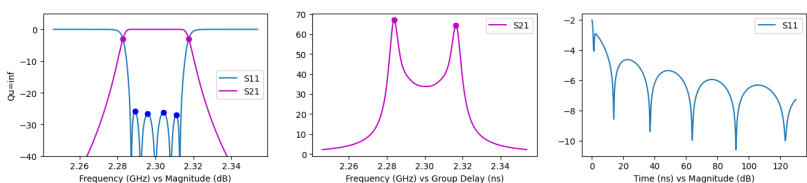
Minimum Return Loss = 26.379 dB
Insertion Loss = 0.000 dB
[0.71287 1.08103781 0.79405141 1.08104169 0.71286771]



5 Pole Chebyshev 0.01 dB

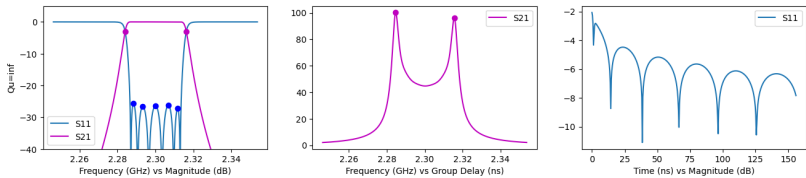
3dB Bandwidth = 34.73675 MHz
Delay Bandwidth = 32.59530 MHz

Transmission Delay = 33.72316 ns
Minimum Return Loss = 26.382 dB
Insertion Loss = 0.000 dB
[0.75633 1.00658949 0.69702688 0.69702688 1.00658949 0.75633]



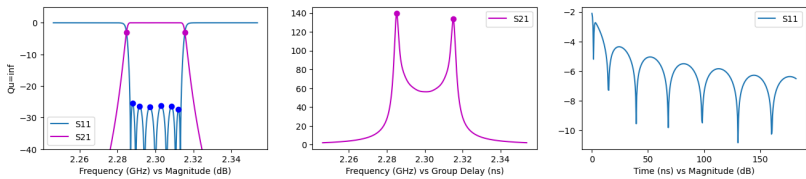
6 Pole Chebyshev 0.01 dB

3dB Bandwidth = 32.27247 MHz
Delay Bandwidth = 30.88429 MHz
Transmission Delay = 44.70244 ns
Minimum Return Loss = 26.381 dB
Insertion Loss = 0.000 dB
[0.78135 0.97007685 0.65967193 0.62092913 0.65967151 0.97007685 0.78135638]



7 Pole Chebyshev 0.01 dB

3dB Bandwidth = 30.80896 MHz
Delay Bandwidth = 29.84046 MHz
Transmission Delay = 56.25512 ns
Minimum Return Loss = 26.374 dB
Insertion Loss = 0.000 dB
[0.79694 0.9492969 0.6409557 0.59183788 0.59183788 0.6409557 0.9492969 0.79694]

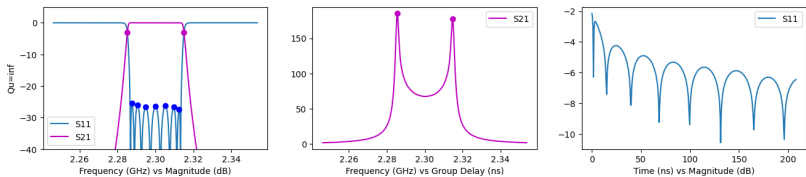


8 Pole Chebyshev 0.01 dB

3dB Bandwidth = 29.88351 MHz
Delay Bandwidth = 29.16252 MHz

Transmission Delay = 67.57205 ns
Minimum Return Loss = 26.382 dB

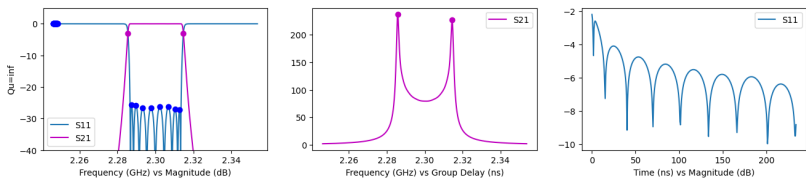
Insertion Loss = 0.000 dB
[0.80728 0.93627737 0.63009954 0.57730673 0.56621917 0.57730673
0.63010033 0.93628028 0.80727904]



9 Pole Chebyshev 0.01 dB

3dB Bandwidth = 29.23784 MHz
Delay Bandwidth = 28.68903 MHz
Transmission Delay = 79.41554 ns
Minimum Return Loss = 0.000 dB

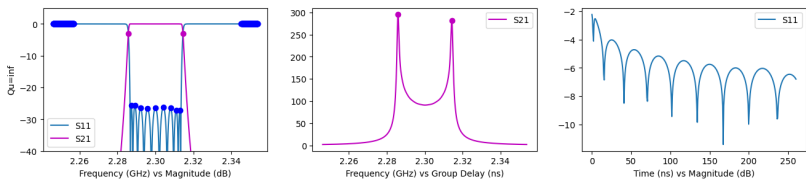
Insertion Loss = 0.000 dB
[0.81446 0.92756425 0.62318543 0.56887656 0.55353121 0.55353121
0.56887656 0.62318543 0.92756425 0.81446]



10 Pole Chebyshev 0.01 dB

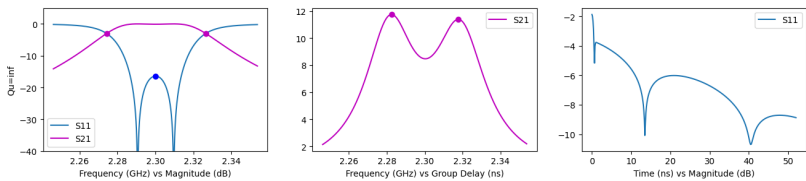
3dB Bandwidth = 28.79664 MHz
Delay Bandwidth = 28.35544 MHz

Transmission Delay = 90.82766 ns
Minimum Return Loss = 0.000 dB
Insertion Loss = 0.000 dB
[0.81965 0.92143231 0.61848657 0.56349365 0.54620785 0.54185
0.54620775 0.56349447 0.61848652 0.92143257 0.81965147]



2 Pole Chebyshev 0.1 dB

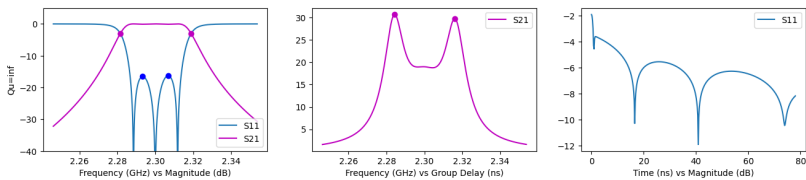
3dB Bandwidth = 52.28807 MHz
Delay Bandwidth = 35.22100 MHz
Transmission Delay = 8.47072 ns
Minimum Return Loss = 16.426 dB
Insertion Loss = 0.000 dB
[0.84304 1.38094768 0.84304747]



3 Pole Chebyshev 0.1 dB

3dB Bandwidth = 37.37322 MHz
Delay Bandwidth = 31.70213 MHz
Transmission Delay = 18.99517 ns
Minimum Return Loss = 16.427 dB

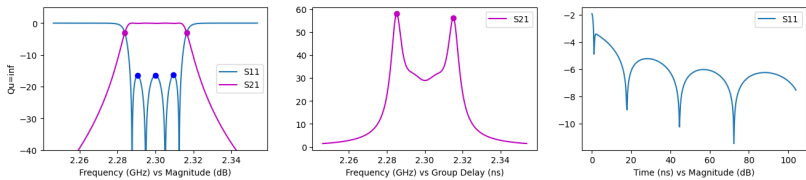
Insertion Loss = 0.000 dB
[1.03156 0.91916892 0.91916892 1.03156]



4 Pole Chebyshev 0.1 dB

3dB Bandwidth = 32.63834 MHz
Delay Bandwidth = 29.82970 MHz
Transmission Delay = 28.92898 ns

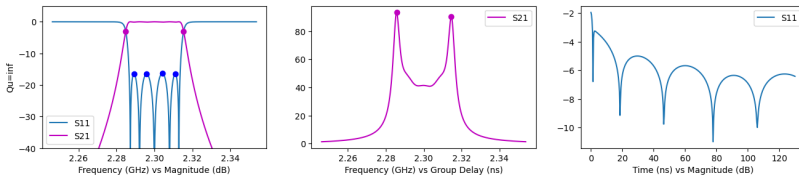
Minimum Return Loss = 16.427 dB
Insertion Loss = 0.000 dB
[1.10879 0.83094798 0.65761077 0.83094551 1.10879291]



5 Pole Chebyshev 0.1 dB

3dB Bandwidth = 30.52917 MHz
Delay Bandwidth = 28.83968 MHz

Transmission Delay = 41.48110 ns
Minimum Return Loss = 16.427 dB
Insertion Loss = 0.000 dB
[1.14681 0.79744783 0.60766534 0.60766534 0.79744783 1.14681]



6 Pole Chebyshev 0.1 dB

3dB Bandwidth = 29.41002 MHz

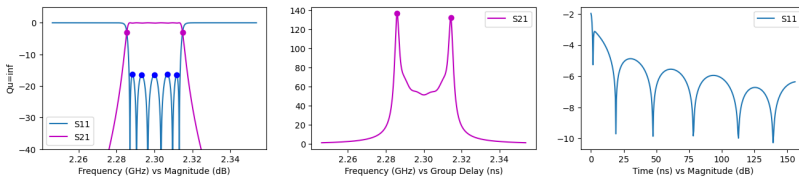
Delay Bandwidth = 28.26935 MHz

Transmission Delay = 51.51704 ns

Minimum Return Loss = 16.428 dB

Insertion Loss = 0.000 dB

[1.16811 0.78087086 0.58855541 0.56618791 0.58855559 0.78087
1.16810346]



7 Pole Chebyshev 0.1 dB

3dB Bandwidth = 28.73207 MHz

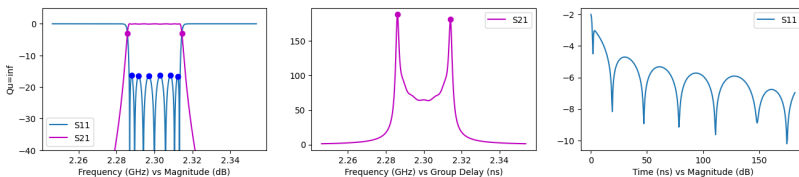
Delay Bandwidth = 27.92499 MHz

Transmission Delay = 64.93239 ns

Minimum Return Loss = 16.428 dB

Insertion Loss = 0.000 dB

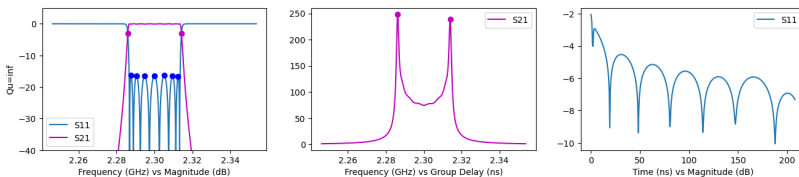
[1.18118 0.77138023 0.57897726 0.5505736 0.5505736 0.578977
0.77138023 1.18118]



8 Pole Chebyshev 0.1 dB

3dB Bandwidth	=	28.30163 MHz
Delay Bandwidth	=	27.68825 MHz
Transmission Delay	=	74.59540 ns
Minimum Return Loss	=	16.428 dB

Insertion Loss = 0.000 dB
 [1.18975 0.76541887 0.57341537 0.54280681 0.53651036 0.54280681 0.57341612 0.76541878 1.18974856]

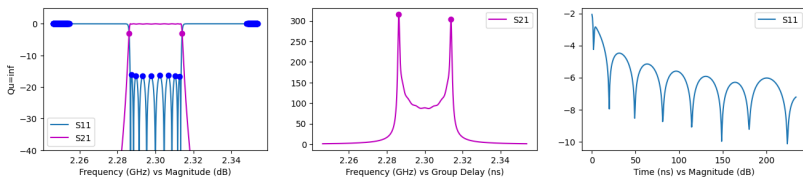


9 Pole Chebyshev 0.1 dB

3dB Bandwidth	=	28.01108 MHz
Delay Bandwidth	=	27.52683 MHz
Transmission Delay	=	88.65610 ns
Minimum Return Loss	=	0.000 dB

Insertion Loss = 0.000 dB
 [1.19567 0.7614149 0.56986747 0.53830626 0.52959255 0.52959255 0.53830626 0.56986747 0.7614149 1.19567]

0.53830626 0.56986747 0.7614149 1.19567]



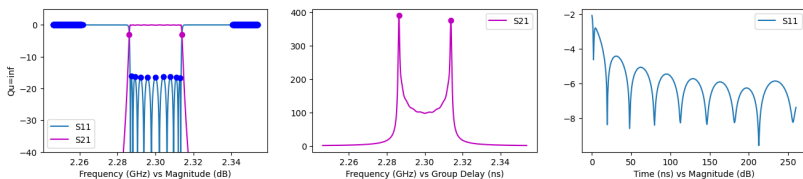
10 Pole Chebyshev 0.1 dB

3dB Bandwidth = 27.79586 MHz
Delay Bandwidth = 27.41922 MHz
Transmission Delay = 97.83672 ns

Minimum Return Loss = 0.000 dB

Insertion Loss = 0.000 dB

[1.19993 0.75859115 0.56745286 0.53543285 0.52561015 0.52315
0.52561015 0.53543137 0.56745227 0.75859245 1.19992732]



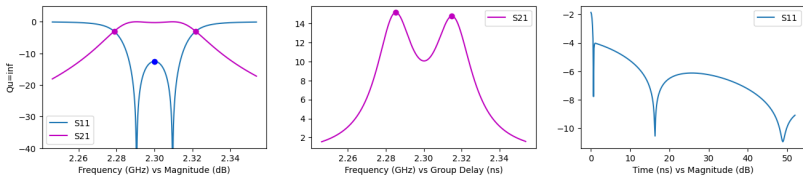
2 Pole Chebyshev 0.25 dB

3dB Bandwidth = 43.00126 MHz
Delay Bandwidth = 29.59296 MHz
Transmission Delay = 10.05671 ns

Minimum Return Loss = 12.522 dB

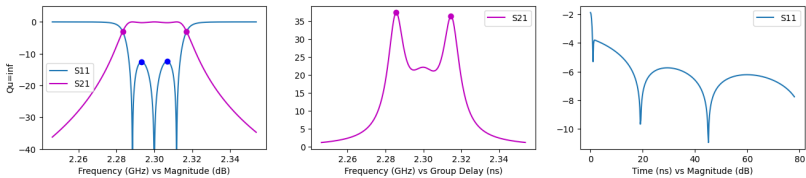
Insertion Loss = 0.000 dB

[1.11316 1.14325132 1.11316285]



3 Pole Chebyshev 0.25 dB

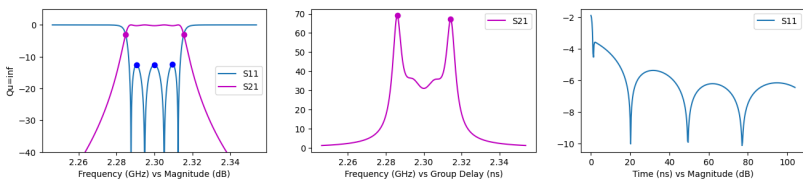
3dB Bandwidth = 33.70369 MHz
 Delay Bandwidth = 28.89349 MHz
 Transmission Delay = 22.20525 ns
 Minimum Return Loss = 12.523 dB
 Insertion Loss = 0.000 dB
 [1.3034 0.81811769 0.81811769 1.3034]



4 Pole Chebyshev 0.25 dB

3dB Bandwidth = 30.66907 MHz
 Delay Bandwidth = 28.18326 MHz
 Transmission Delay = 31.02377 ns

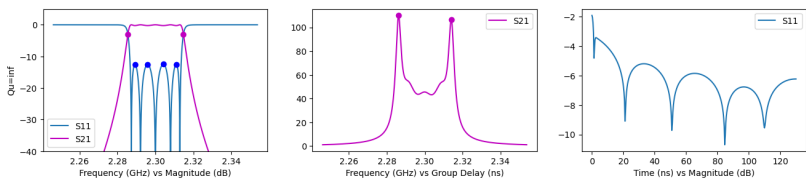
Minimum Return Loss = 12.523 dB
 Insertion Loss = 0.000 dB
 [1.3782 0.7560606 0.61904963 0.75605887 1.37820548]



5 Pole Chebyshev 0.25 dB

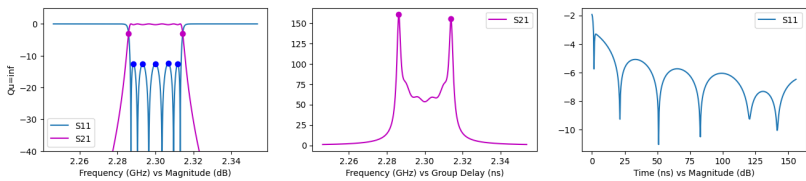
3dB Bandwidth = 29.29165 MHz
Delay Bandwidth = 27.78510 MHz

Transmission Delay = 45.59423 ns
Minimum Return Loss = 12.522 dB
Insertion Loss = 0.000 dB
[1.41442 0.73240526 0.5818136 0.5818136 0.73240526 1.41442



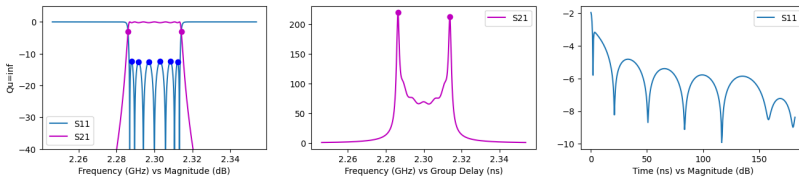
6 Pole Chebyshev 0.25 dB

3dB Bandwidth = 28.54913 MHz
Delay Bandwidth = 27.53759 MHz
Transmission Delay = 53.49130 ns
Minimum Return Loss = 12.523 dB
Insertion Loss = 0.000 dB
[1.43454 0.72067306 0.56760626 0.55030718 0.56760711 0.72067
1.43455032]



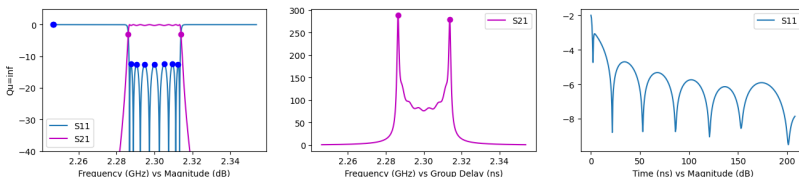
7 Pole Chebyshev 0.25 dB

3dB Bandwidth = 28.10793 MHz
 Delay Bandwidth = 27.36542 MHz
 Transmission Delay = 69.63561 ns
 Minimum Return Loss = 12.523 dB
 Insertion Loss = 0.000 dB
 [1.44683 0.71394475 0.56048631 0.53851035 0.53851035 0.56048631
 0.71394475 1.44683]



8 Pole Chebyshev 0.25 dB

3dB Bandwidth = 27.82814 MHz
 Delay Bandwidth = 27.26857 MHz
 Transmission Delay = 76.28273 ns
 Minimum Return Loss = 12.465 dB
 Insertion Loss = 0.000 dB
 [1.45486 0.70971166 0.55634813 0.53265156 0.52781144 0.53265156
 0.55634871 0.70971394 1.45485973]



9 Pole Chebyshev 0.25 dB

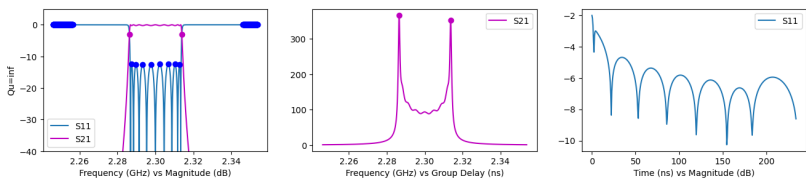
3dB Bandwidth = 27.63444 MHz
 Delay Bandwidth = 27.18248 MHz

Transmission Delay = 93.85463 ns

Minimum Return Loss = 0.000 dB

Insertion Loss = 0.000 dB

[1.4604 0.70686882 0.55371031 0.52925697 0.52255836 0.52255836 0.52925697 0.55371031 0.70686882 1.4604]



10 Pole Chebyshev 0.25 dB

3dB Bandwidth = 27.49455 MHz

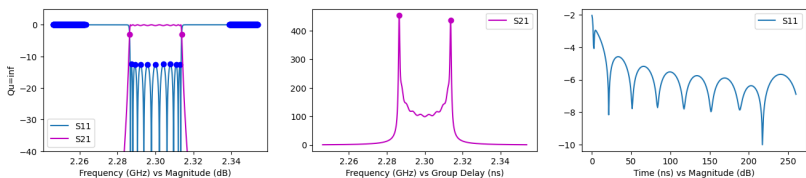
Delay Bandwidth = 27.13943 MHz

Transmission Delay = 99.17951 ns

Minimum Return Loss = 0.000 dB

Insertion Loss = 0.000 dB

[1.46438 0.70486167 0.55191219 0.52708907 0.51953735 0.517651953735 0.51953837 0.52708931 0.55191147 0.70486203 1.4643828]

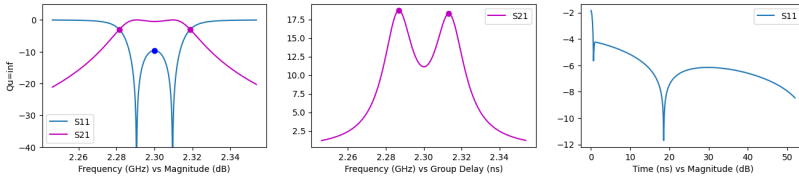


2 Pole Chebyshev 0.5 dB

3dB Bandwidth = 37.38398 MHz

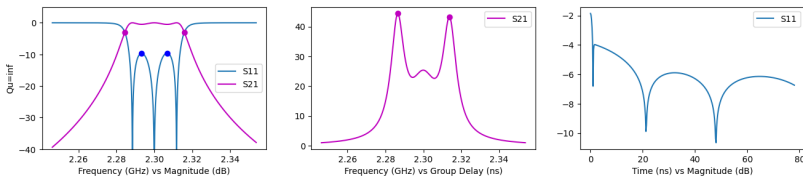
Delay Bandwidth = 26.31083 MHz

Transmission Delay = 11.12612 ns
 Minimum Return Loss = 9.635 dB
 Insertion Loss = 0.000 dB
 [1.40289 1.0040467 1.40288914]



3 Pole Chebyshev 0.5 dB

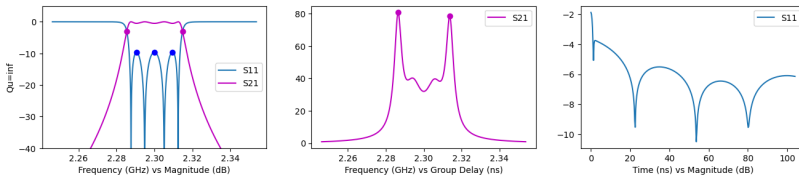
3dB Bandwidth = 31.40082 MHz
 Delay Bandwidth = 27.20400 MHz
 Transmission Delay = 25.37752 ns
 Minimum Return Loss = 9.636 dB
 Insertion Loss = 0.000 dB
 [1.59628 0.75579414 0.75579414 1.59628]



4 Pole Chebyshev 0.5 dB

3dB Bandwidth = 29.41002 MHz
 Delay Bandwidth = 27.19324 MHz
 Transmission Delay = 32.01225 ns
 Minimum Return Loss = 9.636 dB

Insertion Loss = 0.000 dB
 [1.67031 0.70853505 0.59530895 0.7085371 1.67030075]

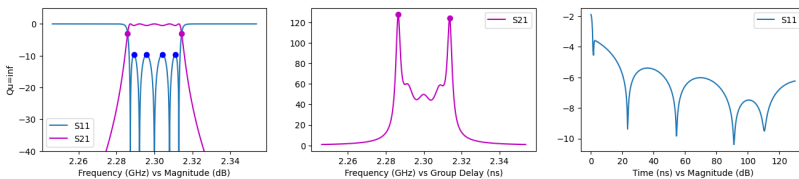


5 Pole Chebyshev 0.5 dB

3dB Bandwidth = 28.49533 MHz
 Delay Bandwidth = 27.11791 MHz

Transmission Delay = 49.76775 ns
 Minimum Return Loss = 9.636 dB
 Insertion Loss = 0.000 dB

[1.70577 0.69048228 0.56575053 0.56575053 0.69048228 1.70577]

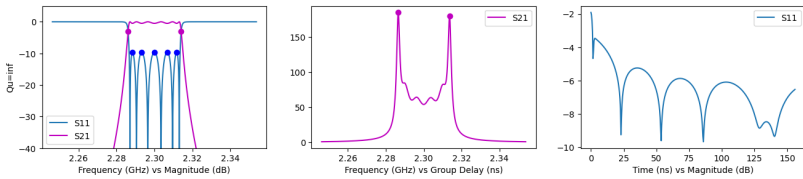


6 Pole Chebyshev 0.5 dB

3dB Bandwidth = 28.00032 MHz
 Delay Bandwidth = 27.07487 MHz

Transmission Delay = 53.98996 ns
 Minimum Return Loss = 9.636 dB
 Insertion Loss = 0.000 dB

[1.72536 0.68151507 0.55449457 0.54043128 0.55449448 0.68151
 1.72535842]



7 Pole Chebyshev 0.5 dB

3dB Bandwidth = 27.70977 MHz

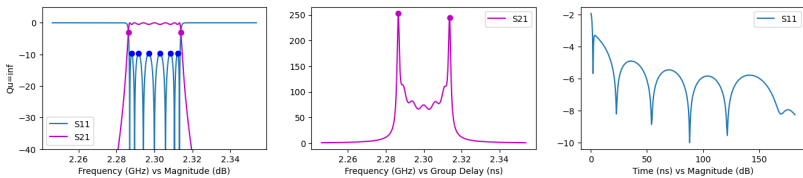
Delay Bandwidth = 27.03182 MHz

Transmission Delay = 74.61923 ns

Minimum Return Loss = 9.635 dB

Insertion Loss = 0.000 dB

[1.73729 0.67636651 0.5488542 0.53098932 0.53098932 0.5488542 0.67636651 1.73729]



8 Pole Chebyshev 0.5 dB

3dB Bandwidth = 27.51607 MHz

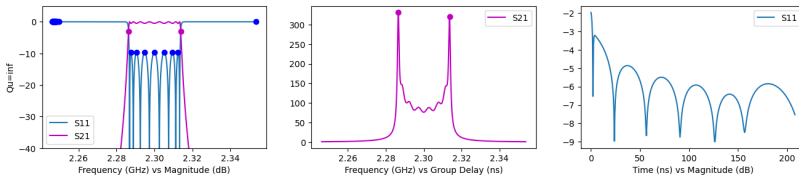
Delay Bandwidth = 27.01030 MHz

Transmission Delay = 76.19326 ns

Minimum Return Loss = 0.000 dB

Insertion Loss = 0.000 dB

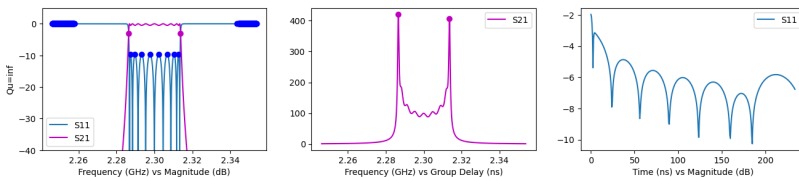
[1.74508 0.67312676 0.54557713 0.52630066 0.52238237 0.52630066 0.54557736 0.67312683 1.74507997]



9 Pole Chebyshev 0.5 dB

3dB Bandwidth = 27.39770 MHz
 Delay Bandwidth = 26.98878 MHz
 Transmission Delay = 99.59360 ns
 Minimum Return Loss = 0.000 dB

Insertion Loss = 0.000 dB
 [1.75044 0.67094779 0.54348472 0.52358636 0.51816461 0.51816461
 0.52358636 0.54348472 0.67094779 1.75044]

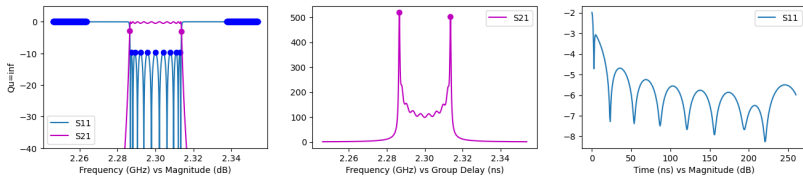


10 Pole Chebyshev 0.5 dB

3dB Bandwidth = 27.29009 MHz
 Delay Bandwidth = 26.96726 MHz
 Transmission Delay = 98.46896 ns

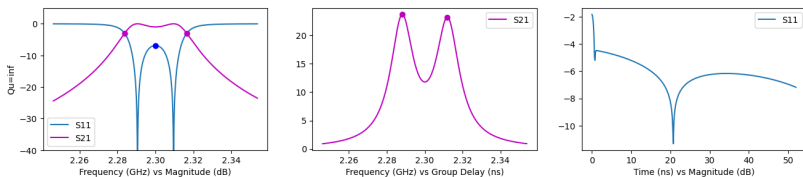
Minimum Return Loss = 0.000 dB
 Insertion Loss = 0.000 dB
 [1.75428 0.66941175 0.54206032 0.52185356 0.51573947 0.51422000
 0.52185356 0.54206032 0.66941175 1.75428]

0.51573942 0.52185323 0.54205907 0.66941101 1.75428601]



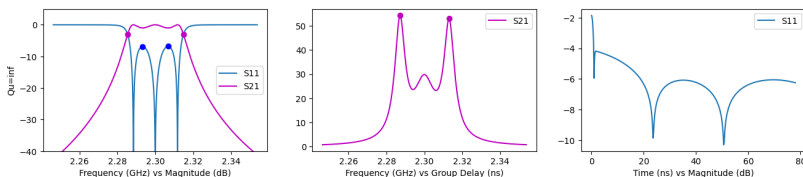
2 Pole Chebyshev 1.0 dB

3dB Bandwidth = 32.75672 MHz
Delay Bandwidth = 23.71741 MHz
Transmission Delay = 11.78182 ns
Minimum Return Loss = 6.868 dB
Insertion Loss = 0.000 dB
[1.82193 0.89512915 1.8219348]



3 Pole Chebyshev 1.0 dB

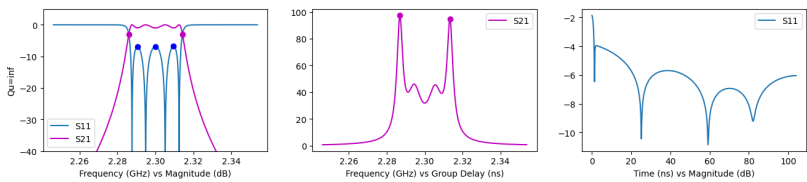
3dB Bandwidth = 29.45307 MHz
Delay Bandwidth = 25.82658 MHz
Transmission Delay = 29.82694 ns
Minimum Return Loss = 6.868 dB
Insertion Loss = 0.000 dB
[2.02359 0.70505614 0.70505614 2.02359]



4 Pole Chebyshev 1.0 dB

3dB Bandwidth = 28.32315 MHz
Delay Bandwidth = 26.37540 MHz
Transmission Delay = 31.88171 ns

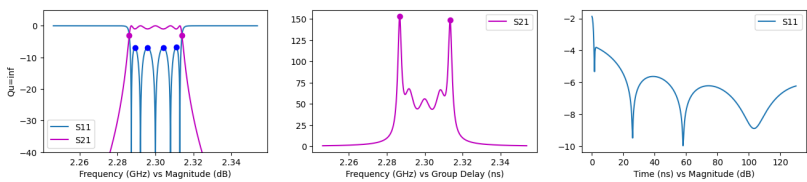
Minimum Return Loss = 6.868 dB
Insertion Loss = 0.000 dB
[2.09905 0.66900292 0.57605012 0.66900186 2.09905102]



5 Pole Chebyshev 1.0 dB

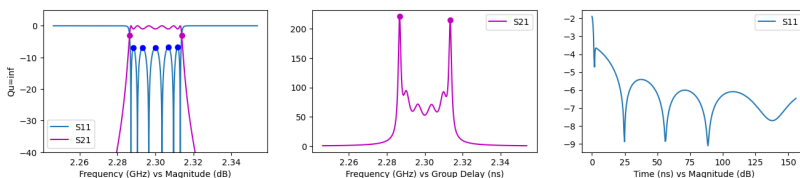
3dB Bandwidth = 27.81738 MHz
Delay Bandwidth = 26.59062 MHz

Transmission Delay = 55.92848 ns
Minimum Return Loss = 6.868 dB
Insertion Loss = 0.000 dB
[2.13488 0.65520771 0.55263517 0.55263517 0.65520771 2.13488]



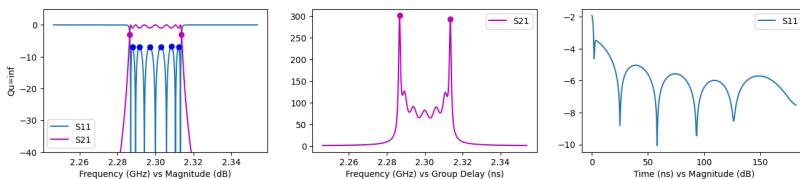
6 Pole Chebyshev 1.0 dB

3dB Bandwidth = 27.53759 MHz
 Delay Bandwidth = 26.69823 MHz
 Transmission Delay = 52.73358 ns
 Minimum Return Loss = 6.868 dB
 Insertion Loss = 0.000 dB
 [2.15459 0.6483471 0.54373417 0.53236802 0.54373361 0.64834
 2.15458598]



7 Pole Chebyshev 1.0 dB

3dB Bandwidth = 27.36542 MHz
 Delay Bandwidth = 26.75204 MHz
 Transmission Delay = 82.34014 ns
 Minimum Return Loss = 6.868 dB
 Insertion Loss = 0.000 dB
 [2.16656 0.64440359 0.53927269 0.52483147 0.52483147 0.53927
 0.64440359 2.16656]

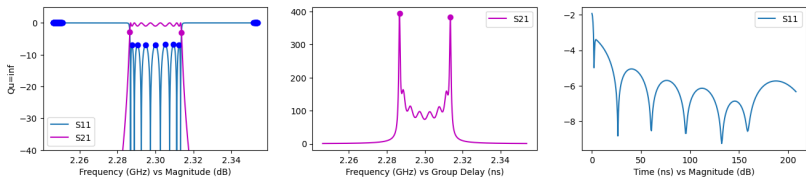


8 Pole Chebyshev 1.0 dB

3dB Bandwidth = 27.24704 MHz

Delay Bandwidth = 26.78432 MHz
Transmission Delay = 73.73537 ns
Minimum Return Loss = 0.000 dB

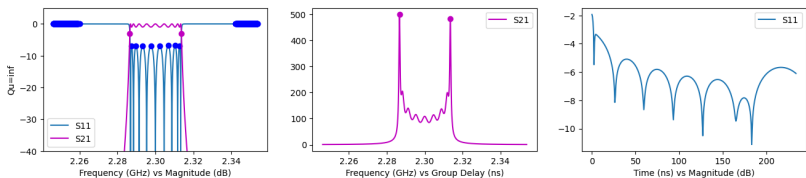
Insertion Loss = 0.000 dB
[2.17436 0.64191981 0.53667905 0.52109408 0.51793852 0.52109408 0.53668003 0.64192163 2.1743477]



9 Pole Chebyshev 1.0 dB

3dB Bandwidth = 27.18248 MHz
Delay Bandwidth = 26.80584 MHz
Transmission Delay = 108.83258 ns
Minimum Return Loss = 0.000 dB

Insertion Loss = 0.000 dB
[2.17972 0.64025013 0.53502396 0.51893139 0.51456493 0.51456493 0.51893139 0.53502396 0.64025013 2.17972]

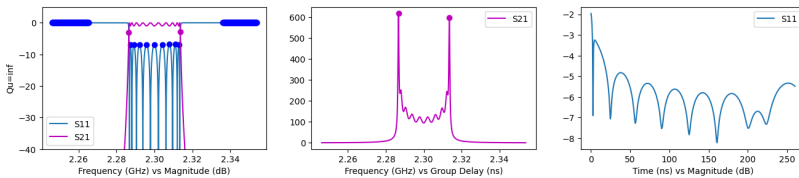


10 Pole Chebyshev 1.0 dB

3dB Bandwidth	=	27.12867 MHz
Delay Bandwidth	=	26.82736 MHz
Transmission Delay	=	94.78460 ns

Minimum Return Loss	=	0.000 dB
Insertion Loss	=	0.000 dB

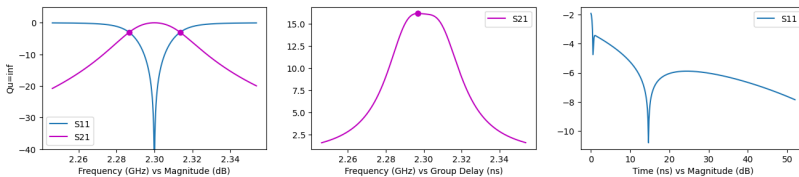
[2.18357 0.6390719 0.5338971 0.51754877 0.51262495 0.511402
0.51262439 0.51754921 0.53389723 0.63907022 2.18357693]



2 Pole Bessel

3dB Bandwidth	=	26.90269 MHz
Delay Bandwidth	=	0.04304 MHz
Transmission Delay	=	16.11251 ns
Minimum Return Loss	=	267.405 dB
Insertion Loss	=	0.000 dB

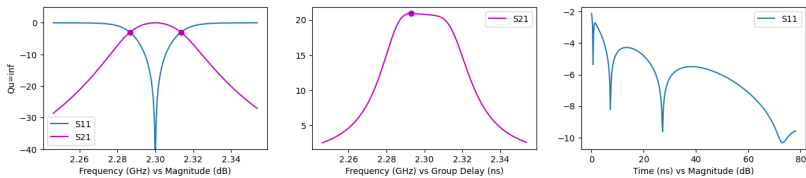
[0.5755 0.89945702 2.1478]



3 Pole Bessel

3dB Bandwidth	=	26.89193 MHz
Delay Bandwidth	=	0.03228 MHz
Transmission Delay	=	20.77475 ns

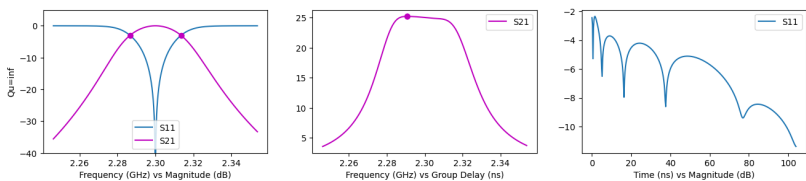
Minimum Return Loss = 260.046 dB
 Insertion Loss = 0.000 dB
 [0.3374 1.74755032 0.68384166 2.2034]



4 Pole Bessel

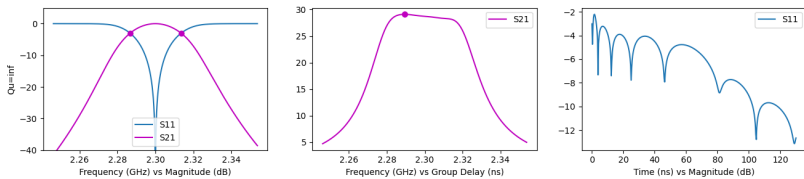
3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 0.00000 MHz
 Transmission Delay = 25.01395 ns

Minimum Return Loss = 279.423 dB
 Insertion Loss = 0.000 dB
 [0.2334 2.52408183 1.1725749 0.64242725 2.2404]



5 Pole Bessel

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 0.03228 MHz
 Transmission Delay = 28.72303 ns
 Minimum Return Loss = 262.753 dB
 Insertion Loss = 0.000 dB
 [0.1743 3.36326962 1.56596609 1.05807133 0.63133777 2.2582



6 Pole Bessel

3dB Bandwidth = 26.90269 MHz

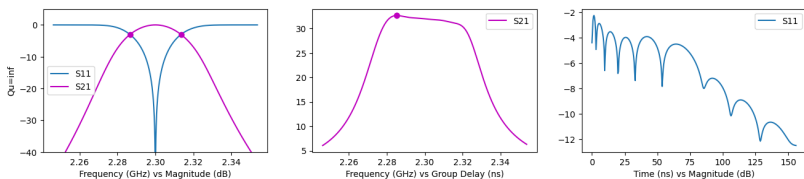
Delay Bandwidth = 0.00000 MHz

Transmission Delay = 31.98955 ns

Minimum Return Loss = 260.683 dB

Insertion Loss = 0.000 dB

[0.1365 4.27853542 1.97716573 1.35364113 1.0260119 0.630005
2.2645]



7 Pole Bessel

3dB Bandwidth = 26.90269 MHz

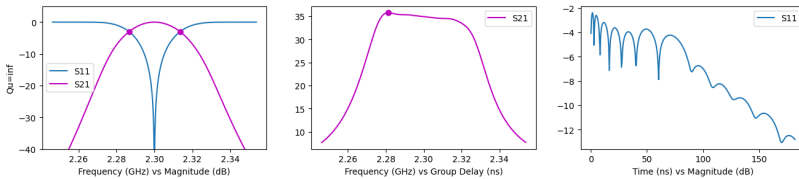
Delay Bandwidth = 0.00000 MHz

Transmission Delay = 34.92829 ns

Minimum Return Loss = 266.908 dB

Insertion Loss = 0.000 dB

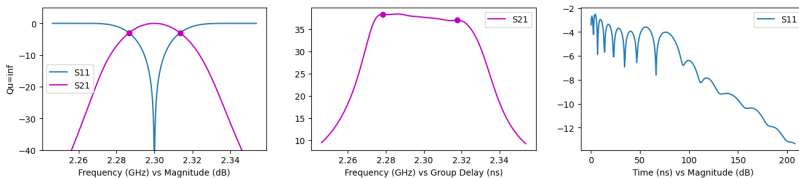
[0.1106 5.26720542 2.41779532 1.6473776 1.28032892 1.020398
0.63191577 2.2659]



8 Pole Bessel

3dB Bandwidth	=	26.89193 MHz
Delay Bandwidth	=	39.43934 MHz
Transmission Delay	=	37.62506 ns
Minimum Return Loss	=	254.503 dB

Insertion Loss = 0.000 dB
 [0.0919 6.32612313 2.88818725 1.95471301 1.51880635 1.25491
 1.02456424 0.63472027 2.2656]

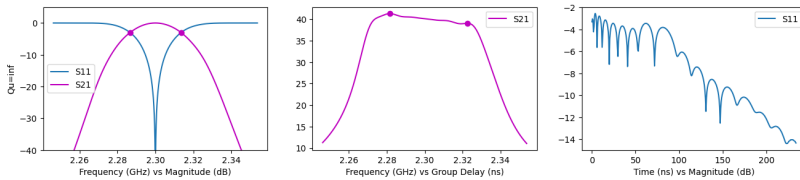


9 Pole Bessel

3dB Bandwidth	=	26.90269 MHz
Delay Bandwidth	=	40.93513 MHz
Transmission Delay	=	40.13486 ns

Minimum Return Loss = 260.582 dB
 Insertion Loss = 0.000 dB
 [0.078 7.44500309 3.38642537 2.27878777 1.7619662 1.463193

1.25010672 1.03227016 0.63752995 2.2649]



10 Pole Bessel

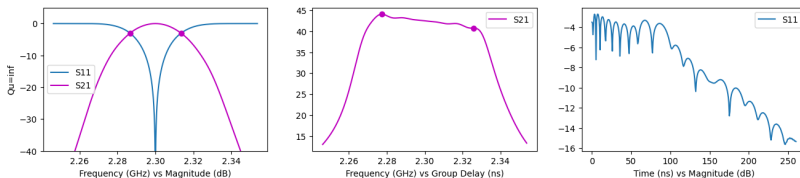
3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 48.38180 MHz

Transmission Delay = 42.49141 ns

Minimum Return Loss = 272.623 dB

Insertion Loss = 0.000 dB

[0.0672 8.63013564 3.91226606 2.6203 2.01530544 1.6691419
1.44070594 1.25468816 1.04089813 0.64006293 2.2641]



2 Pole Linear Phase 0.05 Deg

3dB Bandwidth = 26.90269 MHz

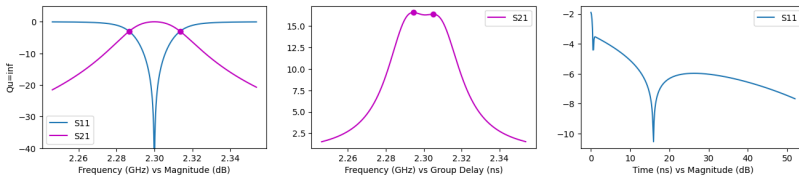
Delay Bandwidth = 10.42748 MHz

Transmission Delay = 16.30894 ns

Minimum Return Loss = 268.989 dB

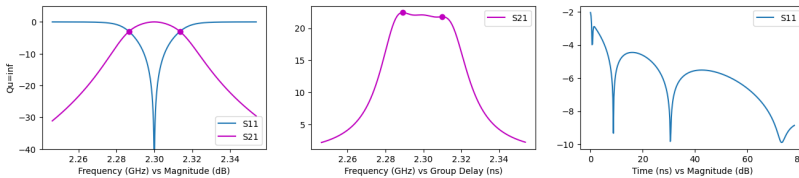
Insertion Loss = 0.000 dB

[0.648 0.85551119 2.1085]



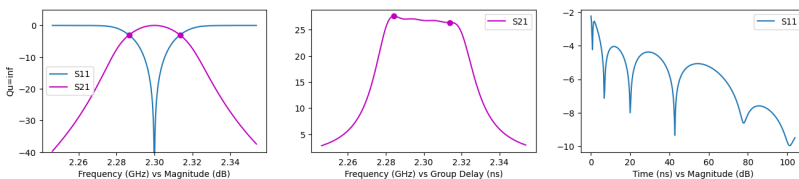
3 Pole Linear Phase 0.05 Deg

3dB Bandwidth = 26.89193 MHz
 Delay Bandwidth = 20.86573 MHz
 Transmission Delay = 22.06692 ns
 Minimum Return Loss = 261.030 dB
 Insertion Loss = 0.000 dB
 [0.4328 1.48859545 0.65226502 2.2542]



4 Pole Linear Phase 0.05 Deg

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 29.61448 MHz
 Transmission Delay = 26.75046 ns
 Minimum Return Loss = 259.740 dB
 Insertion Loss = 0.000 dB
 [0.3363 1.93240525 1.04827813 0.62419412 2.2459]



5 Pole Linear Phase 0.05 Deg

3dB Bandwidth = 26.90269 MHz

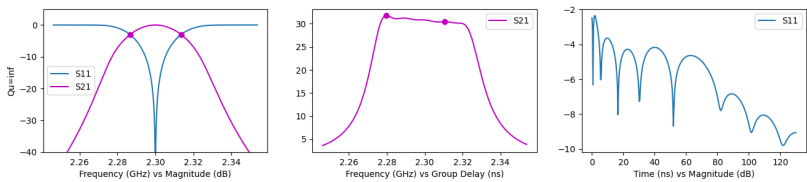
Delay Bandwidth = 30.80896 MHz

Transmission Delay = 30.81985 ns

Minimum Return Loss = 269.903 dB

Insertion Loss = 0.000 dB

[0.2751 2.35739648 1.31122795 1.00956383 0.62946552 2.2873



6 Pole Linear Phase 0.05 Deg

3dB Bandwidth = 26.90269 MHz

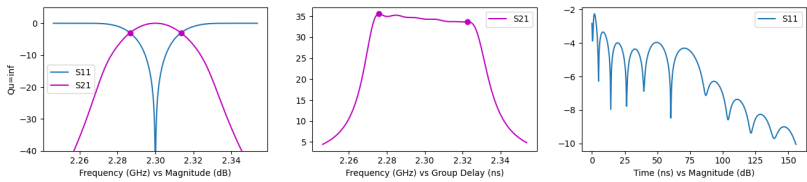
Delay Bandwidth = 46.61698 MHz

Transmission Delay = 34.30706 ns

Minimum Return Loss = 258.619 dB

Insertion Loss = 0.000 dB

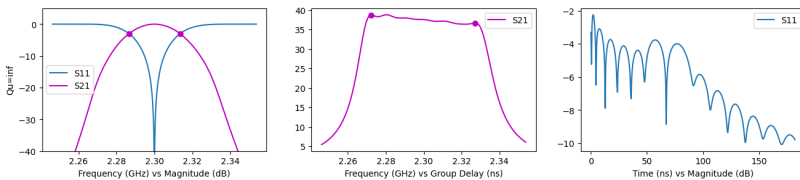
[0.2374 2.7275625 1.52664397 1.22735627 1.01124755 0.631714
2.2448]



7 Pole Linear Phase 0.05 Deg

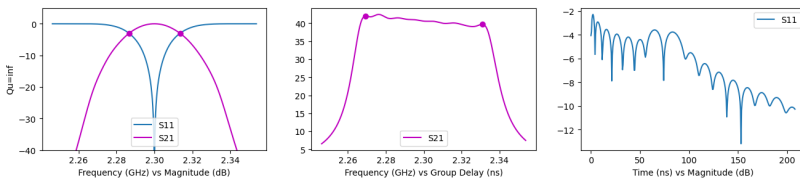
3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 54.69855 MHz

Transmission Delay = 37.58364 ns
 Minimum Return Loss = 271.478 dB
 Insertion Loss = 0.000 dB
 [0.2085 3.09745783 1.7340023 1.41368758 1.23277267 1.034947
 0.64047452 2.2845]



8 Pole Linear Phase 0.05 Deg

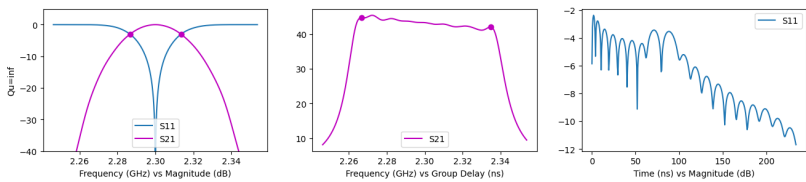
3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 61.49955 MHz
 Transmission Delay = 40.55670 ns
 Minimum Return Loss = 259.585 dB
 Insertion Loss = 0.000 dB
 [0.1891 3.41179452 1.91044248 1.56730475 1.39709914 1.25038
 1.04335075 0.63973109 2.2415]



9 Pole Linear Phase 0.05 Deg

3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 68.07457 MHz
Transmission Delay = 43.33925 ns
Minimum Return Loss = 260.522 dB

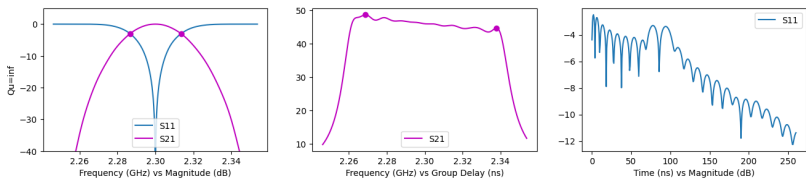
Insertion Loss = 0.000 dB
[0.1718 3.7469144 2.09451263 1.72225145 1.55158771 1.426752
1.27744658 1.06432888 0.64746039 2.2834]



10 Pole Linear Phase 0.05 Deg

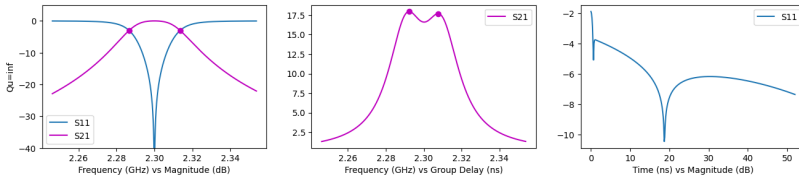
3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 69.08611 MHz
Transmission Delay = 45.99874 ns

Minimum Return Loss = 258.607 dB
Insertion Loss = 0.000 dB
[0.1601 4.01899298 2.2462908 1.84986254 1.67606063 1.562970
1.44351578 1.29255959 1.06568558 0.64410207 2.2387]



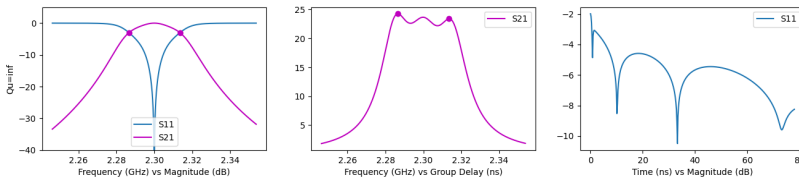
2 Pole Linear Phase 0.5 Deg

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 15.22692 MHz
 Transmission Delay = 16.59294 ns
 Minimum Return Loss = 260.689 dB
 Insertion Loss = 0.000 dB
 [0.8245 0.78265808 1.98]



3 Pole Linear Phase 0.5 Deg

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 27.06411 MHz
 Transmission Delay = 23.66735 ns
 Minimum Return Loss = 255.364 dB
 Insertion Loss = 0.000 dB
 [0.5534 1.32983413 0.63527414 2.425]

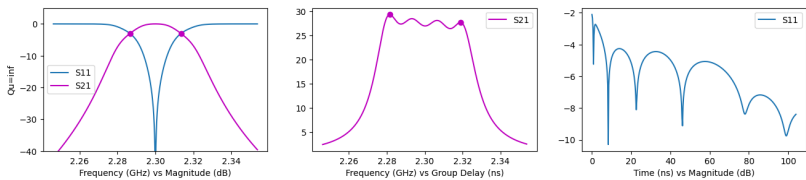


4 Pole Linear Phase 0.5 Deg

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 37.25485 MHz
 Transmission Delay = 27.01847 ns

Minimum Return Loss = 257.625 dB

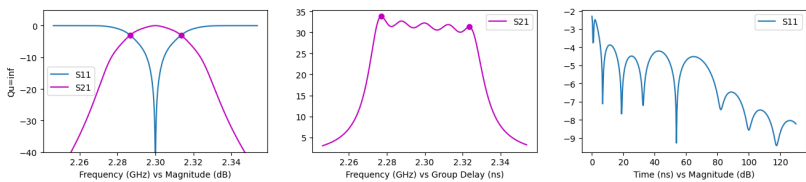
Insertion Loss = 0.000 dB
[0.4526 1.66531087 0.99536284 0.62045392 2.0504]



5 Pole Linear Phase 0.5 Deg

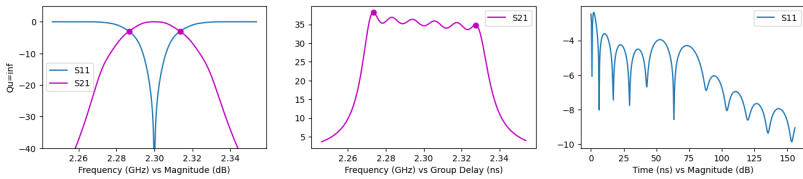
3dB Bandwidth = 26.90269 MHz

Delay Bandwidth = 46.31567 MHz
Transmission Delay = 32.24396 ns
Minimum Return Loss = 252.731 dB
Insertion Loss = 0.000 dB
[0.3658 2.00977747 1.24626695 1.01953291 0.6359985 2.4446



6 Pole Linear Phase 0.5 Deg

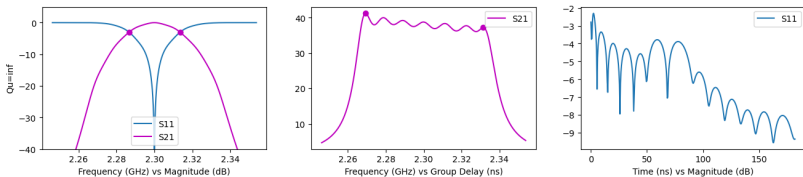
3dB Bandwidth = 26.89193 MHz
Delay Bandwidth = 53.88071 MHz
Transmission Delay = 34.60940 ns
Minimum Return Loss = 253.354 dB
Insertion Loss = 0.000 dB
[0.3313 2.2459159 1.41131263 1.22335751 0.99300618 0.624885
2.0111]



7 Pole Linear Phase 0.5 Deg

3dB Bandwidth	=	26.90269 MHz
Delay Bandwidth	=	62.09141 MHz
Transmission Delay	=	38.75984 ns

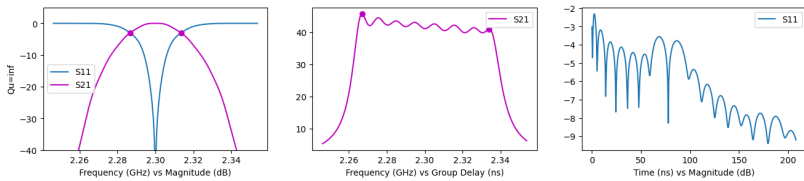
Minimum Return Loss = 260.689 dB
Insertion Loss = 0.000 dB
[0.2826 2.57613638 1.62048497 1.41551224 1.24589558 1.06297
0.65333192 2.4404]



8 Pole Linear Phase 0.5 Deg

3dB Bandwidth	=	26.90269 MHz
Delay Bandwidth	=	67.08455 MHz
Transmission Delay	=	41.16374 ns
Minimum Return Loss	=	260.202 dB
Insertion Loss	=	0.000 dB

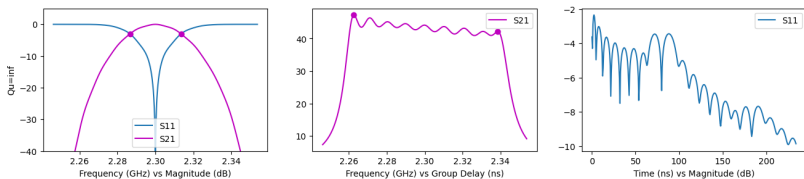
[0.2718 2.7128995 1.71515738 1.52637783 1.36539448 1.257045
1.00960268 0.62190675 1.9626]



9 Pole Linear Phase 0.5 Deg

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 75.85483 MHz
 Transmission Delay = 44.14391 ns

Minimum Return Loss = 264.900 dB
 Insertion Loss = 0.000 dB
 [0.2347 3.07946481 1.93995265 1.71529468 1.57360028 1.47376
 1.30631273 1.09912198 0.66638143 2.4332]

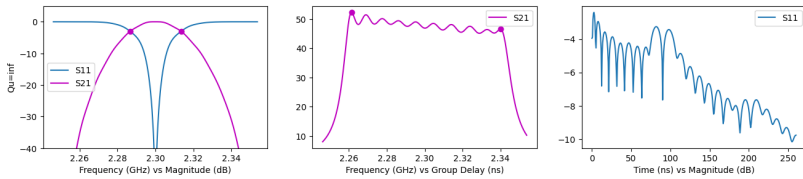


10 Pole Linear Phase 0.5 Deg

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 78.50205 MHz
 Transmission Delay = 46.88327 ns

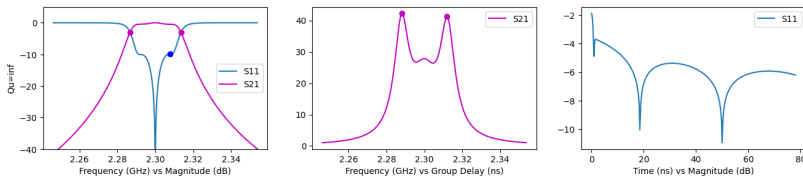
Minimum Return Loss = 276.916 dB
 Insertion Loss = 0.000 dB
 [0.2359 3.11490644 1.9717959 1.7690213 1.61837402 1.561983

1.40287588 1.29209665 1.00838698 0.61560131 1.8431]



3 Pole Legendre

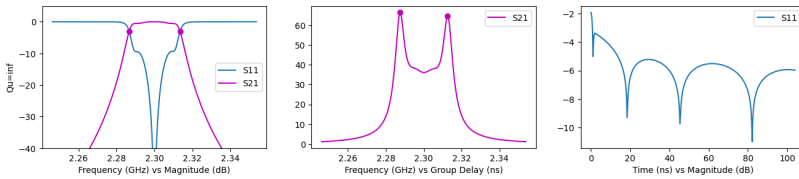
3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 23.72817 MHz
Transmission Delay = 27.85270 ns
Minimum Return Loss = 9.940 dB
Insertion Loss = 0.000 dB
[1.1737 0.79331231 0.58208249 2.1801]



4 Pole Legendre

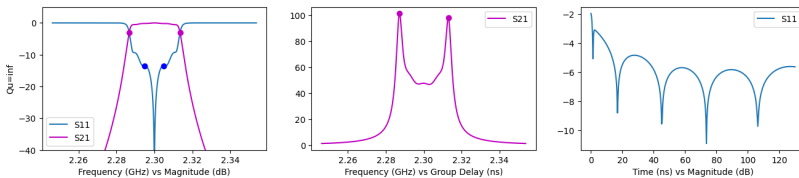
3dB Bandwidth = 26.82736 MHz
Delay Bandwidth = 25.05179 MHz
Transmission Delay = 36.11811 ns

Minimum Return Loss = 267.405 dB
Insertion Loss = 0.000 dB
[1.0828 0.79023538 0.58759744 0.56767946 1.5845]



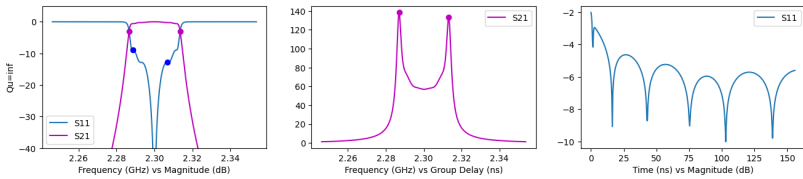
5 Pole Legendre

3dB Bandwidth = 26.80584 MHz
 Delay Bandwidth = 25.71897 MHz
 Transmission Delay = 47.65774 ns
 Minimum Return Loss = 13.535 dB
 Insertion Loss = 0.000 dB
 [0.9512 0.84338692 0.56933823 0.55785036 0.57003793 1.999



6 Pole Legendre

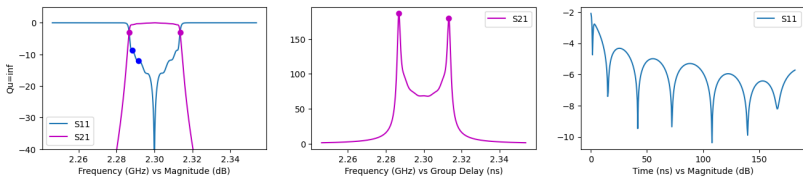
3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 26.16018 MHz
 Transmission Delay = 56.88991 ns
 Minimum Return Loss = 12.746 dB
 Insertion Loss = 0.000 dB
 [0.918 0.85641879 0.58230526 0.53733489 0.54874223 0.576861
 1.5783]



7 Pole Legendre

3dB Bandwidth = 26.87041 MHz
 Delay Bandwidth = 26.36464 MHz
 Transmission Delay = 68.67387 ns

Minimum Return Loss = 10.301 dB
 Insertion Loss = 0.000 dB
 [0.8394 0.89810119 0.59084839 0.54641188 0.51886433 0.54084
 0.57786685 1.884]

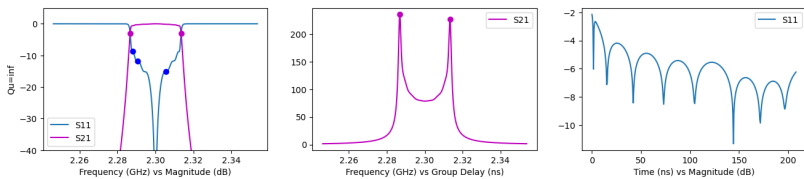


8 Pole Legendre

3dB Bandwidth = 26.90269 MHz
 Delay Bandwidth = 26.49377 MHz
 Transmission Delay = 78.49535 ns
 Minimum Return Loss = 13.466 dB

Insertion Loss = 0.000 dB
 [0.8205 0.91091819 0.59680371 0.54408906 0.52519597 0.51454

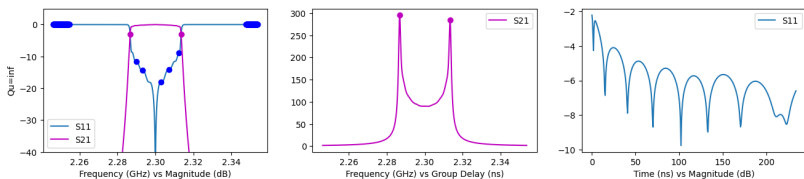
0.54183083 0.58930725 1.5564]



9 Pole Legendre

3dB Bandwidth = 26.90269 MHz
Delay Bandwidth = 26.57986 MHz
Transmission Delay = 90.23968 ns

Minimum Return Loss = 0.000 dB
Insertion Loss = 0.000 dB
[0.7695 0.94490812 0.60656203 0.54918793 0.52209975 0.52120
0.50993976 0.53586198 0.59267709 1.7645]

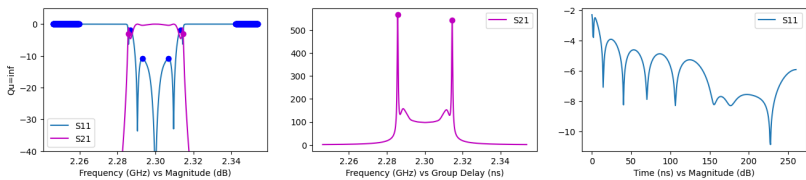


10 Pole Legendre

3dB Bandwidth = 29.01186 MHz
Delay Bandwidth = 28.75360 MHz
Transmission Delay = 97.59628 ns

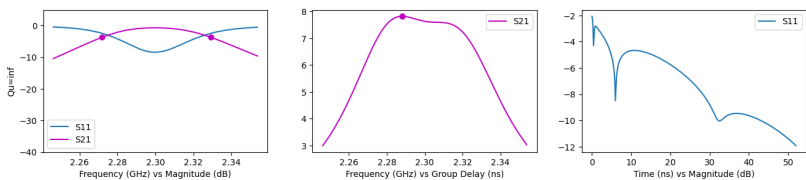
Minimum Return Loss = 0.000 dB

Insertion Loss = 0.000 dB
[0.7575 0.95568475 0.61092268 0.54991431 0.52514363 0.60471
0.60350318 0.50845223 0.53958262 0.60082779 1.5286]



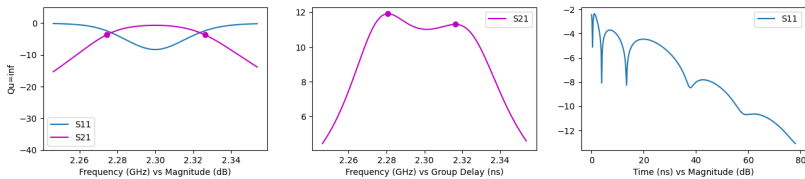
2 Pole Gaussian

3dB Bandwidth = 57.41034 MHz
Delay Bandwidth = 0.02152 MHz
Transmission Delay = 7.63101 ns
Minimum Return Loss = 8.403 dB
Insertion Loss = 0.000 dB
[0.2624 2.16016441 1.81813754]



3 Pole Gaussian

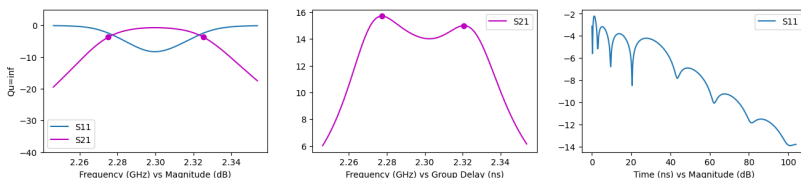
3dB Bandwidth = 51.94371 MHz
Delay Bandwidth = 35.45775 MHz
Transmission Delay = 11.01472 ns
Minimum Return Loss = 8.321 dB
Insertion Loss = 0.000 dB
[0.1772 3.26248317 1.42248889 2.0925645]



4 Pole Gaussian

3dB Bandwidth = 50.23270 MHz
 Delay Bandwidth = 43.31333 MHz
 Transmission Delay = 14.04622 ns

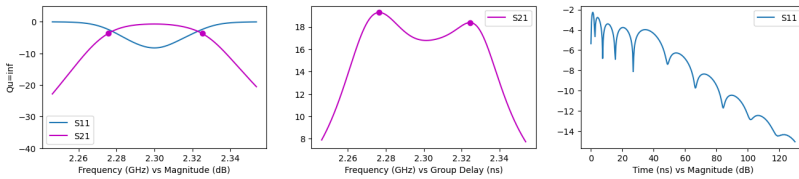
Minimum Return Loss = 8.285 dB
 Insertion Loss = 0.000 dB
 [0.1312 4.42306792 1.98946148 1.25554181 2.20417806]



5 Pole Gaussian

3dB Bandwidth = 49.65161 MHz

Delay Bandwidth = 47.85451 MHz
 Transmission Delay = 16.79728 ns
 Minimum Return Loss = 8.270 dB
 Insertion Loss = 0.000 dB
 [0.1026 5.65760563 2.56181463 1.68363052 1.192056 2.252737]



6 Pole Gaussian

3dB Bandwidth = 49.43638 MHz

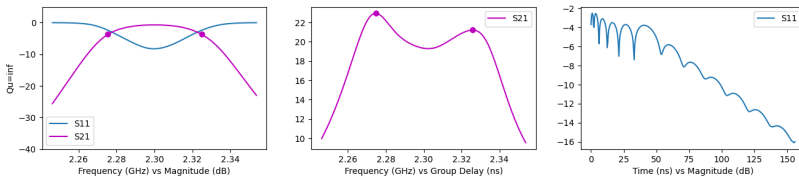
Delay Bandwidth = 51.13663 MHz

Transmission Delay = 19.32547 ns

Minimum Return Loss = 8.264 dB

Insertion Loss = 0.000 dB

[0.0833 6.96731492 3.15785739 2.09738435 1.55966961 1.16353
2.27478559]



7 Pole Gaussian

3dB Bandwidth = 49.38258 MHz

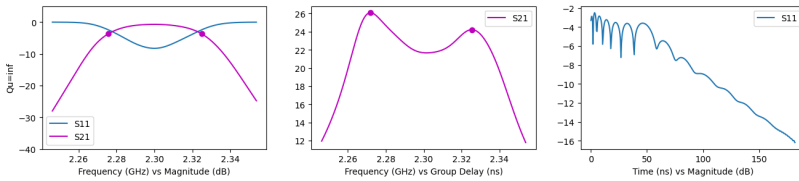
Delay Bandwidth = 53.42874 MHz

Transmission Delay = 21.67662 ns

Minimum Return Loss = 8.261 dB

Insertion Loss = 0.000 dB

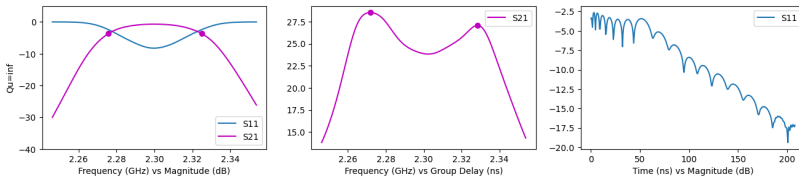
[0.0695 8.34732974 3.78066565 2.5172628 1.90078911 1.50007
1.14967165 2.2852044]



8 Pole Gaussian

3dB Bandwidth = 49.36106 MHz
 Delay Bandwidth = 56.57098 MHz
 Transmission Delay = 23.88178 ns
 Minimum Return Loss = 8.260 dB

Insertion Loss = 0.000 dB
 [0.0591 9.80226821 4.43119822 2.95013285 2.23806494 1.80119
 1.46886587 1.14261337 2.29025241]

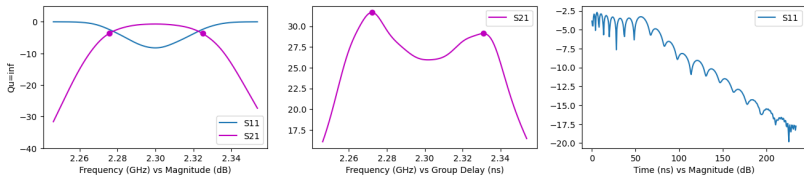


9 Pole Gaussian

3dB Bandwidth = 49.35030 MHz
 Delay Bandwidth = 58.73395 MHz
 Transmission Delay = 25.96411 ns

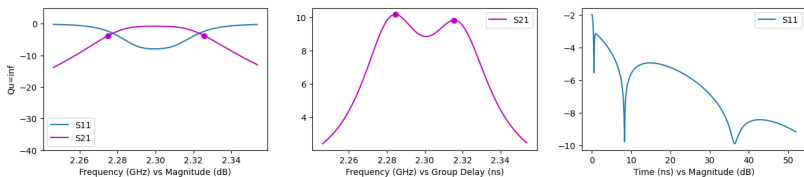
Minimum Return Loss = 8.259 dB
 Insertion Loss = 0.000 dB
 [0.0512 11.31696826 5.11228136 3.39842109 2.58008009 2.0

1.74658168 1.45193605 1.13896429 2.29261318]



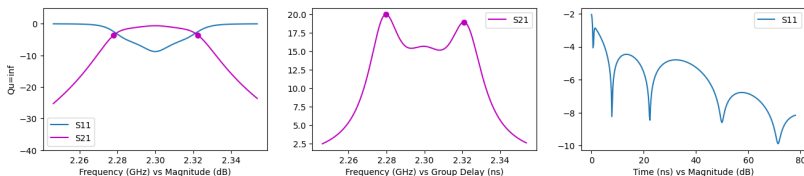
2 Pole Gaussian 6 dB

3dB Bandwidth = 50.72771 MHz
 Delay Bandwidth = 31.06723 MHz
 Transmission Delay = 8.85489 ns
 Minimum Return Loss = 7.941 dB
 Insertion Loss = 0.000 dB
 [0.4042 1.66214555 2.093679]



3 Pole Gaussian 6 dB

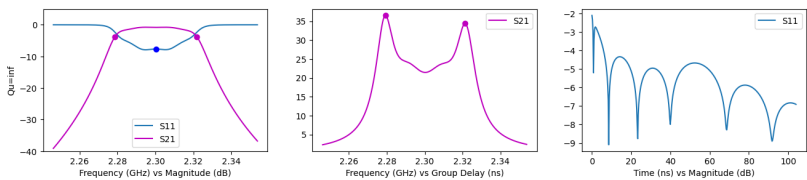
3dB Bandwidth = 44.52933 MHz
 Delay Bandwidth = 41.10731 MHz
 Transmission Delay = 15.69288 ns
 Minimum Return Loss = 8.787 dB
 Insertion Loss = 0.000 dB
 [0.4198 1.74398379 1.0492334 2.48510346]



4 Pole Gaussian 6 dB

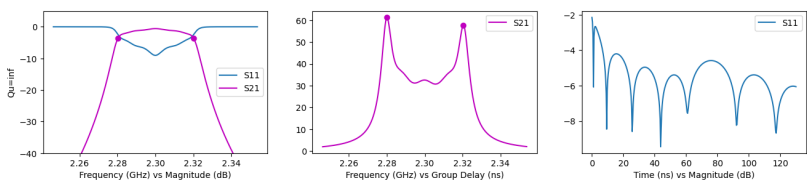
3dB Bandwidth = 43.18420 MHz
Delay Bandwidth = 41.97896 MHz
Transmission Delay = 21.45185 ns

Minimum Return Loss = 7.656 dB
Insertion Loss = 0.000 dB
[0.4544 1.61314228 1.04040155 0.92176597 2.60062812]



5 Pole Gaussian 6 dB

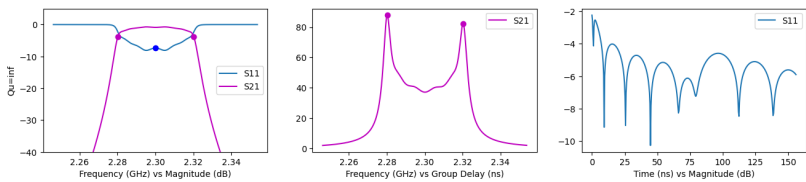
3dB Bandwidth = 39.80522 MHz
Delay Bandwidth = 40.30023 MHz
Transmission Delay = 32.59751 ns
Minimum Return Loss = 9.042 dB
Insertion Loss = 0.000 dB
[0.5041 1.48200506 0.95424327 0.88786404 0.82123566 2.97272]



6 Pole Gaussian 6 dB

3dB Bandwidth = 39.96664 MHz
Delay Bandwidth = 39.99892 MHz
Transmission Delay = 37.14821 ns
Minimum Return Loss = 7.346 dB

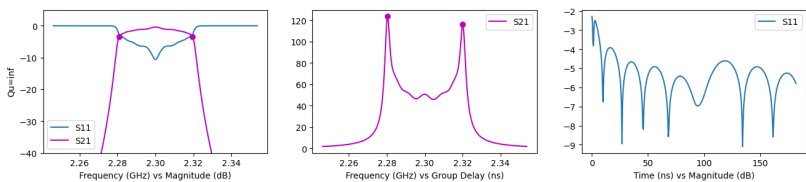
Insertion Loss = 0.000 dB
[0.4918 1.4840818 0.94435539 0.85646483 0.82299511 0.8163602.85619873]



7 Pole Gaussian 6 dB

3dB Bandwidth = 38.79368 MHz
Delay Bandwidth = 39.65457 MHz

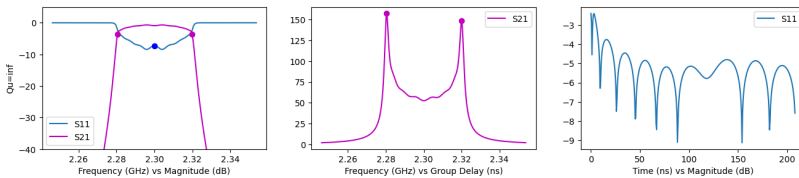
Transmission Delay = 50.80955 ns
Minimum Return Loss = 10.608 dB
Insertion Loss = 0.000 dB
[0.52835562 1.43155936 0.91484768 0.84666643 0.78688244 0.8190.77339729 3.0856]



8 Pole Gaussian 6 dB

3dB Bandwidth	=	39.29945 MHz
Delay Bandwidth	=	39.52543 MHz
Transmission Delay	=	52.50140 ns
Minimum Return Loss	=	7.269 dB

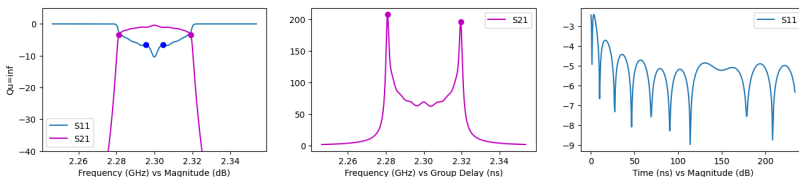
Insertion Loss = 0.000 dB
[0.4979 1.46429533 0.92896069 0.83516948 0.78946074 0.78671
0.78282447 0.79127737 2.88713894]



9 Pole Gaussian 6 dB

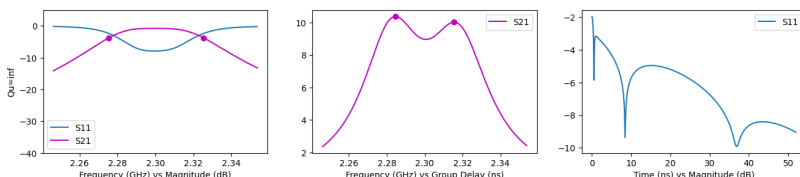
3dB Bandwidth	=	37.98660 MHz
Delay Bandwidth	=	38.59998 MHz
Transmission Delay	=	69.01065 ns

Minimum Return Loss = 6.603 dB
Insertion Loss = 0.000 dB
[0.53243704 1.40374973 0.89506498 0.81757552 0.76096942 0.768
0.74283142 0.77439455 0.75288228 3.19224]



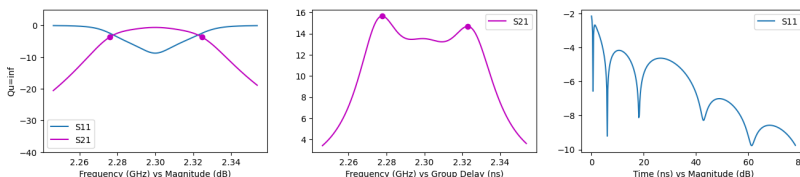
2 Pole Gaussian 12 dB

3dB Bandwidth = 50.06053 MHz
 Delay Bandwidth = 30.88429 MHz
 Transmission Delay = 8.97635 ns
 Minimum Return Loss = 7.913 dB
 Insertion Loss = 0.000 dB
 [0.4152 1.63134959 2.122406]



3 Pole Gaussian 12 dB

3dB Bandwidth = 48.47865 MHz
 Delay Bandwidth = 45.11043 MHz
 Transmission Delay = 13.52102 ns
 Minimum Return Loss = 8.744 dB
 Insertion Loss = 0.000 dB
 [0.3097 2.22113304 1.20069642 2.28047764]

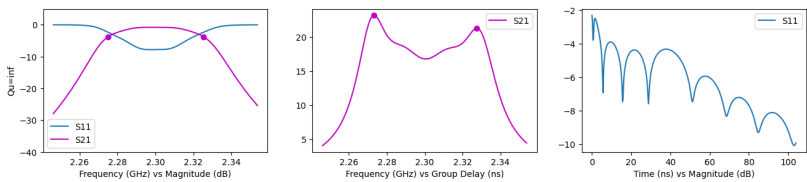


4 Pole Gaussian 12 dB

3dB Bandwidth = 50.38336 MHz
 Delay Bandwidth = 54.19278 MHz
 Transmission Delay = 16.76816 ns

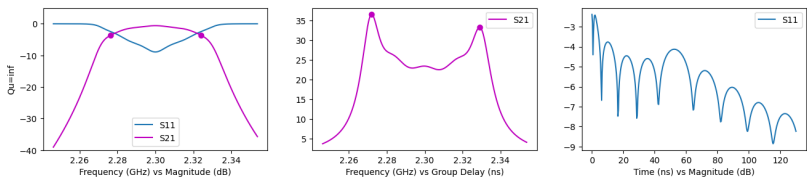
Minimum Return Loss = 7.801 dB

Insertion Loss = 0.000 dB
[0.2909 2.42679612 1.45325285 1.12965926 2.293767]



5 Pole Gaussian 12 dB

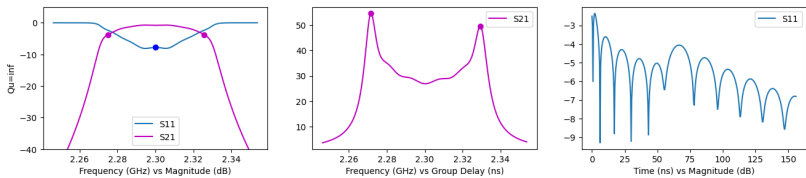
3dB Bandwidth = 47.55320 MHz
Delay Bandwidth = 56.91533 MHz
Transmission Delay = 23.43638 ns
Minimum Return Loss = 8.922 dB
Insertion Loss = 0.000 dB
[0.3164 2.28185171 1.43844933 1.26248377 1.05286353 2.42170]



6 Pole Gaussian 12 dB

3dB Bandwidth = 50.66315 MHz
Delay Bandwidth = 57.59328 MHz
Transmission Delay = 26.91287 ns
Minimum Return Loss = 7.769 dB

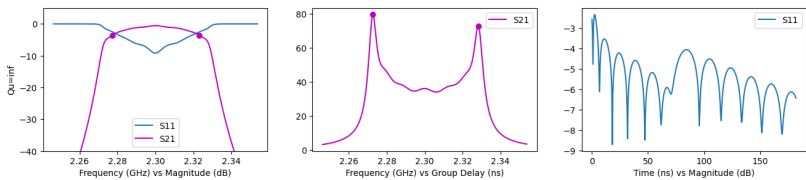
Insertion Loss = 0.000 dB
 [0.3207 2.23059473 1.40432961 1.26259483 1.18142355 1.05900
 2.29926021]



7 Pole Gaussian 12 dB

 3dB Bandwidth = 45.79914 MHz
 Delay Bandwidth = 55.53791 MHz
 Transmission Delay = 36.16355 ns

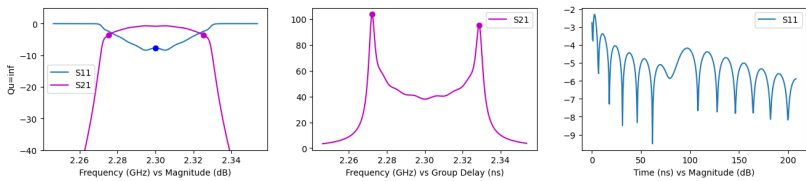
Minimum Return Loss = 9.202 dB
 Insertion Loss = 0.000 dB
 [0.3449 2.10153215 1.32425836 1.19753001 1.13315173 1.12243
 0.98869684 2.57711836]



8 Pole Gaussian 12 dB

 3dB Bandwidth = 49.98520 MHz
 Delay Bandwidth = 56.42032 MHz
 Transmission Delay = 37.99130 ns
 Minimum Return Loss = 7.656 dB

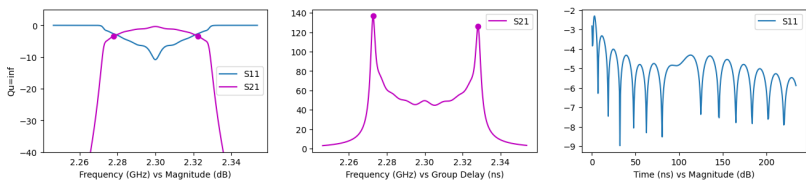
Insertion Loss = 0.000 dB
[0.3318 2.15330293 1.34796477 1.20255221 1.13973794 1.13474
1.10966112 1.02639488 2.324682]



9 Pole Gaussian 12 dB

3dB Bandwidth = 44.36792 MHz
Delay Bandwidth = 55.34421 MHz
Transmission Delay = 49.46486 ns

Minimum Return Loss = 10.827 dB
Insertion Loss = 0.000 dB
[0.354865 2.0653488 1.30126877 1.17981542 1.09783385 1.11149
1.0728859 1.10394555 0.9649933 2.62059566]



In []: